TERM 3

NETWORKS AND INTERNET TECHNOLOGIES

CHAPTER 7

CHAPTER OVERVIEW

By the end of this chapter, you will be able to:

- Describe local area networks (LANs)
- Discuss the advantages and disadvantages of LANs
- Describe wireless local area networks (WLANs)
- Describe the basic components of a network
- Explain various types of connections
- Discuss the basic components of a network, including workstations, servers and network devices
- Describe how network connection speed is measured on wired and wireless networks
- Explain how wired and wireless network communication works
- Discuss the differences between the internet and an intranet
- Define passwords in terms of basic network security
- Discuss what usernames are
- Describe what access rights are in terms of networks on web services
- Discuss the advantages and disadvantages of various forms of online communication

INTRODUCTION

In Grade 10, you learned about home area networks (HANs) and personal area networks (PANs). HANs are very small networks that usually cover a single home. PANs, on the other hand, are much smaller and are usually designed to serve a single user.

In this section, you will learn about local area networks and wide area networks and how they are generally used.

Something to know

In 1973, the Defence Advanced Research Projects Agency (DARPA) in the United States began researching the techniques and technologies needed to develop communication protocols that would allow computers in the same network to communicate with each other across multiple linked networks. This was called the "Internetting" project and it resulted in what we know as the internet today. This was also the first functioning example of LANs communicating with each other. Before this, computers could only communicate with each other if they were connected in the same network.

NETWORKS AND INTERNET TECHNOLOGIES

What is the difference between the internet, a network, and an intranet?

https://www.youtube.com/watch?v=nojwX3_XZBs

UNIT

7.1 Networks

In this unit, you will be focusing on:

- LANs and WLANs
- Basic network components
- Network software
- Network connections
- Network communication

TYPES OF NETWORKS

LOCAL AREA NETWORKS

A **local area network (LAN)** is a small network of computers covering a small area, such as an office building or school. A **wireless local area network (WLAN)** is the same as a LAN but it has the ability to connect wireless devices such as smartphones, laptops and tablets to the LAN.

LANs may serve only one or two users (for example in a home) or they can serve hundreds of users (in an office building or on a school campus). No matter how many users LANs serve, they are all designed to share resources such as internet connections, printers or server connections. Most LANs use either wireless or wired connections or a combination of the two to connect devices. For example, desktop computers and laptops can be connected to the network with cables while the printer and mobile devices are connected using wireless connections.

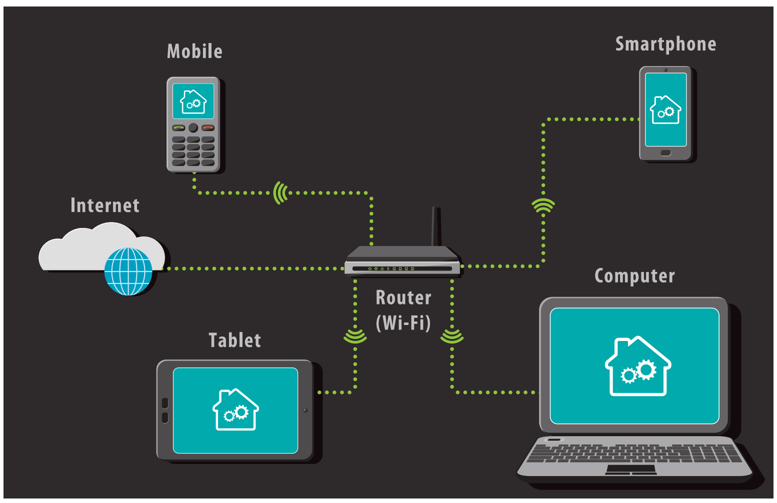

Figure 7.1: *A LAN in which computers are connected in the same room*

Gr11_CAT_Theory_LB.indb 92 2019/11/19 10:35 AM

LANs have several advantages and disadvantages.

Some of the advantages of LAN are:

- Software (and licenses), files and hardware can be shared easily with the devices connected to the network.
- Files and other data can be shared faster if they remain on the network and do not need to be uploaded to the internet or emailed.
- Networks can be centrally controlled, making it easier to make changes, monitor users, update software, troubleshoot hardware and software issues and maintain resources.

Some of the disadvantages of LAN are:

- The initial setup costs for creating a network can be high, especially in a school or office environment, as you need to make sure that you have enough resources for the number of computers.
- There is a risk of privacy and data violations. **Network administrators** may have access to all the files created by each user.
- If a network is not secured properly, one infected computer can infect the entire network.

WIRELESS LAN (WLAN)

A wireless LAN (WLAN) is a wireless computer network that links two or more devices using wireless communication to form a local area network (LAN) within a limited area such as a home, school, computer laboratory, campus, office building etc. This gives users the ability to move around within the area and yet still be connected to the network. A WLAN allows users to move around the coverage area, often a home or small office, while maintaining a network connection.

BASIC COMPONENTS OF A NETWORK

A network consists of the following basic components:

- Workstation and servers
- NIC
- Network devices
- Communication media
- Network software

WORKSTATIONS, CLIENTS AND SERVERS

Workstations and servers are usually part of a network. Workstations are the computers connected to the network. They are used by people to carry out tasks, for example the accountant who creates spreadsheets or the designer who designs websites.

Workstations will have input and output devices because they are designed to be used by humans. Some workstations, such as those designed to run **computer-aided design (CAD)** applications, will be more powerful than the computers you can buy to use at home.

Figure 7.2: *Workstations in a computer laboratory*

Servers are designed to manage network resources and hardly ever carry out tasks beyond their server tasks (for example, you would not use a server to design an advertisement).

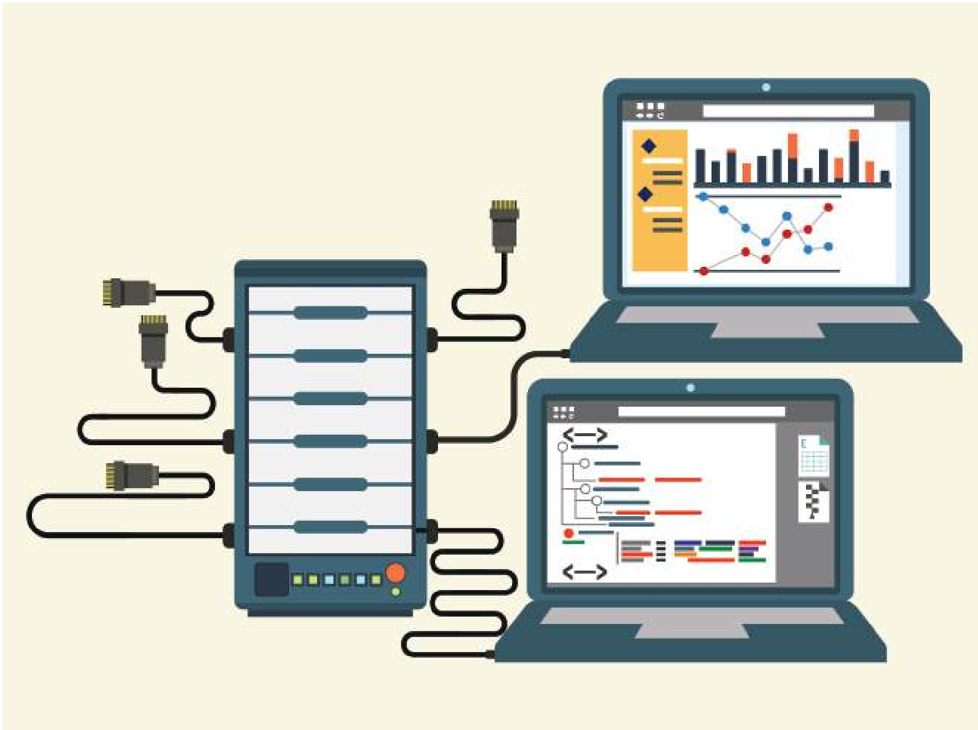

Figure 7.3: *A computer server connected to laptops*

The main function of a server is to serve the information stored on it to other computers that are connected to it by a network. A server will usually have more RAM than a normal computer so that it can process data faster and it will use special operating systems, such as Linux and Windows Server, to carry out its tasks. Servers also usually have faster CPUs and larger hard drives, and they are often connected to UPSs.

Servers are usually only used for one type of task, for example:

- Email servers send and receive emails and store each user's email information (their email address, username and password). Email servers can be local, such as those used by businesses, or global, such as Gmail. You interact with email servers through an email client, such as Microsoft Outlook or Gmail.
- Web servers are where the World Wide Web is. You will use a web browser, such as Google Chrome or Internet Explorer, to interact with a web server. Web servers deliver web pages to you when you request them, but they are also where you can upload data to cloud storage systems or your own website.
- Database servers are usually local servers that cannot be accessed by users outside the network they are connected to. Users can access database servers using specialised software (such as Microsoft SQL) to interact with them.
- A file server is a computer responsible for the central storage and management of data files so that other computers on the same network can access the files.

Some networks have a dedicated server which serves workstation (called clients). This is known as a client-server network. Other networks have workstations (called nodes) which act as both server and client. This is known as a peer-to-peer network.

NETWORK INTERFACE CARDS

Network interface cards (NICs) are pieces of hardware that allow a computer to connect to a network. Most modern computers have this device integrated into their motherboards, but NICs can also be purchased separately.

Figure 7.4: *A network interface card*

Network interface cards are also called network interface controllers or network cards. An NIC connects using **ethernet** cables, while a wireless NIC (WNIC) uses an antenna to connect to a wireless network. Smartphones use WNICs to connect to data signals and Wi-Fi hotspots. Laptops and desktops can use both NICs and WNICs to connect to a network.

Something to know

Network cards that plug into a computer or laptop using a USB cable are not NICs. They are called network adapters.

CONNECTION DEVICES

For a network to function, it needs some of the following:

SWITCHES	Switches, which are small devices that act as a controller for the networked devices to communicate (they create the network).	
ROUTERS	Routers, which allow multiple computers to connect to the network, but not necessarily the internet (they guide the traffic on the network).	
MODEMS	Modems, which connect to an internet service provider (ISP) to give internet access. Connecting the router to the modem gives the network internet access	
ACCESS POINTS	Access points are devices that are used to set up a WLAN in a large building such as an office or school. They allow wireless devices such as smartphones to connect to the network.	

Most modern routers combine the modem and the switches. A modem is a communication device. Modems are hardware devices that connect a computer or router to a broadband internet network. They work by converting the digital data sent by the computer into the **analogue signal** used by telephone lines. Modems are classified by how much data they can send (in bits per second) and how they connect.

Early modems used a dial-up connection to communicate but now they use digital connections (digital subscriber line, or DSL), fibre or wireless.

NETWORK SOFTWARE

Network software refers to a range of software aimed at the design, implementation and operation of computer networks. It exposes the inner workings of the network to the network administrators, assisting them in managing and monitoring the network. It also allows multiple devices such as desktops, laptops, tablets, mobile phones and other systems to connect to each other as well as to other networks.

It is important that you understand the difference between network software and software applications. Network software is mostly used by network administrators, while software applications allow users who are working within a network to do their work.

If your network has a server, that server will need a server operating system. You can choose between different types of server operating systems to suit your needs. Device management software will allow you to manage all the devices connected to a network.

The functions of network software can be summarised as follows.

- Its main function is to set up and install computer networks.
- It allows network administrators to add or remove users from a network.
- It helps administrators to protect the network from data breaches, unauthorised access and attacks on a network through the use of a security tool.
- It helps administrators to define locations of data storage and allows users to access that data.

COMMUNICATION MEDIA

Network connections can be wired or wireless. Wired network connections use ethernet cables to connect all the devices in a network, such as the computers, routers and switches. Ethernet cables are made up of several twisted pairs of wires inside a plastic casing and have a connector on either end called an RJ45 connector, which plugs into network ports on the various devices.

ETHERNET VS WI-FI

https://www.youtube.com/watch?v=l6wfN4Y7dOU

Figure 7.5: *An ethernet cable with RJ45 connector*

Wireless networks use radio signals to connect all the devices in a network (such as the computers, routers and printers). When you connect to a Wi-Fi hotspot in a restaurant or hotel, you are connecting to a wireless network.

HOW FAST IS MY NETWORK?

Wired and wireless networks can only transfer data as fast as their connections allow. The speed at which a network can transfer data is based on its components.

WIRED SPEEDS

For wired networks, the speed of the network is determined by the rated speed of the cables you use. The most common cable speed is **1 Gb (gigabit)**, these cables can transfer 1 000 Mb data per second.

WIRELESS SPEEDS

The speed of wireless networks depends on the standards they use. Wi-Fi standards are certified by the Institute of Electrical and Electronics Engineers (IEEE). The main IEEE standard for wireless networks is 802.11 and there are a number of specifications under this banner.

The 802.11 standards are different in terms of speed, transmission ranges and the frequency they use, but there are only a few of these standards that you will need to know. These standards make it easy for you to know what speeds you can expect from a given **wireless access point** (WAP).

LOOKING AT ADVERTISEMENTS FOR NETWORK DEVICES

Now that you have a better understanding of what a LAN is and how they are created, you will be able to take what you have learned and use it to interpret advertisements for network devices. There are several key pieces of information you will need to look out for when it comes to deciding which networking devices you want to buy.

Look at the rated speed for NICs and cables and match them to make sure that your data is transferred as quickly as possible. Make sure that WNICs are using the newest wireless standard (802.11ac).

On routers, check the number of ports, whether it is able to handle wireless connections, how many networks it can connect and whether or not it has a built-in modem.

Lastly, look at how well these devices meet your requirements and how suitable a connection type is to the space available (work out if a wired or wireless connection would be better).

Something to know

Remember: The frequency of the bandwidth determines how much data can be carried through a specific network at any given time. A one GHz connection is equal to a 1 000 MHz connection.

NETWORK COMMUNICATION

Now that you understand how networks are connected, you need to understand how the different components of a network communicate with each other. Wired and wireless networks use very different communication media to connect the various components.

WIRED COMMUNICATION

There are two types of cables that connect modern wired networks to each other, namely unshielded twisted pair (UTP) cables and fibre optic cables. UTP cables are commonly used to connect LANs and telephone networks as they are easy to make and set up and are relatively cost-effective.

COPPER VS FIBRE

https://www.youtube.com/watch?v=_Bw2NFBDxR8

They are made up of two cables that are twisted together (the twisted pair) to cancel out electromagnetic interference from outside sources. They are called unshielded because no extra interference shields, such as metal meshes or aluminium foil, are added to the cables.

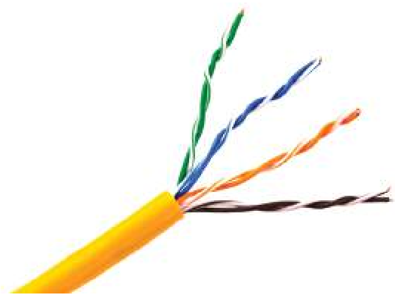

Figure 7.6: *UTP cable with different twist rates*

The wires are then connected to a connector (such as the RJ45 connector of an ethernet cable).

Fibre optic cables, on the other hand, are made of thousands of very thin strands of pure glass, called optical fibres, inside an insulating jacket. Instead of using electrical signals, fibre optic cables use light to transmit data very quickly.

Figure 7.7: *A multi-fibre cable*

Fibre optic cables are largely replacing metal cables since they are more flexible and lighter and can transmit data over longer distances with lower rates of data loss. The most common application for fibre optic cables is in internet connections.

WIRELESS COMMUNICATION

As you know, wireless connections allow you to transfer data over a distance without using cables, wires or physical connectors. Most common wireless communication uses radio waves to send data from one point to another.

A transmitter on the sending end turns the signal into an electromagnetic radio wave. The antenna picks up the radio wave and the receiver turns the electrical signal back into data.

Activity 7.1

Answer the following questions:

1. Briefly describe the difference between a workstation and a server.
2. List two functions of network software.
3. How do wireless networks connect devices in a network?
4. Provide two advantages of using fibre optic cable to connect to a network.

UNIT

7.2 Intranet vs internet

While they may sound like the same thing, there are actually a fairly large number of differences between the internet and an intranet. In this unit, you will learn about these differences as well as what an extranet and a virtual private network (VPN) are.

INTRANET VS INTERNET

An intranet is a private, web-based network that is only available to an organisation's staff. Intranets work using **internet protocols** to share data and information across the network. To put it simply, an intranet is like a private internet. Very often, a company's intranet will have information on it that the company does not want the public to have access to, such as employee contact details and calendars.

Intranets are also a valuable communication tool in a company as they allow all members of staff to work together, find information, create content and share tasks quickly and easily. Intranets act as a website and a communication channel together. Access to an intranet is limited to users who have been given usernames and **passwords** to connect with it.

The internet, on the other hand, is a massive network of networks with millions of computers connected to each other across the globe. All of these computers can communicate with each other as long as they are connected. By design, the internet is decentralised, meaning that it is not controlled by a single, central authority. Each computer, or host, is independent and the owner or operator of that host can choose which internet services to use and which local services they will make publicly available.

The internet itself is publicly accessible and is not the World Wide Web (WWW), which is simply a way to access information over the internet. Access to the internet is usually provided by ISPs.

INTRANET VS EXTRANET

https://www.youtube.com/watch?v=zlEstGNVdwo

HOW A VPN WORKS

https://www.youtube.com/watch?v=rFg7TSwVcL4

> VPNs are groups of computers (a network) connected over a public network (like the internet). Businesses use VPNs to allow their staff to connect to their internal networks without them being connected to the LAN (for example, if a staff member has to work in an office in Cape Town, they can connect to the main office in Johannesburg using a VPN).
>
> VPNs can also be used to secure your computer's internet connection to keep the data you are sending and receiving safe from unwanted access when you are using an internet connection you do not trust (like a Wi-Fi connection in a coffee shop). A VPN encrypts all the data your computer sends over an internet connection.

Activity 7.2

1. Answer the following questions in your own words.
   a. Define an intranet.
   b. What are the main differences between an intranet and the internet?
   c. What is the main purpose of an extranet?
   d. What would you use a VPN for?

UNIT

7.3 Basic network security

In our modern age, more and more of our important and personal data is being stored digitally and hackers and other criminals want to get hold of that information. That is why it is extremely important that the networks we use daily are secured and protected.

In this unit, you will learn about the basics of passwords, usernames and access rights, and how those three factors combine to secure a network.

PASSWORDS

Passwords are the most basic form of network security. Passwords, at their core, are a secret string of letters, numbers and symbols created by users or generated by a computer to limit or restrict access to a system.

More often than not, you will need to pick your own password to make sure that it is something you can remember.

USERNAMES

Usernames are a unique identifier given to any person who uses a secure computer network. Usernames are also called an account name, login ID or user ID. Usernames are most commonly used with passwords and can be used on computers or websites.

ACCESS RIGHTS

On a computer, for example, a user may only have access to a personal folder where they can open, read, write, create and delete files and folders. A user with administrative rights can access all the files and folders on a computer and can make changes that will affect all users.

Activity 7.3

Answer the following in your own words.

1. Read the following case study and answer the questions which follow:

> **Cyber-attacks: Protecting universities and solving cyber security issues**
>
> Oxford, Warwick, and Greenwich Universities are among many of the higher education institutes to have fallen victim to attacks in recent years, with hackers attempting to steal research data and documents. The problem has become acute enough to warrant the publication of cybersecurity guidance for universities and colleges by the UK Government's National Cyber Security Centre (NCSC) recently.
>
> No organisation today is immune from the threat of cyber-attacks. While we most often hear about breaches suffered by retailers, financial institutions and, since the infamous WannaCry ransomware attack in 2017, healthcare providers, many other types of organisations are becoming appealing targets for cybercriminals.

… continued

Activity 7.3 *… continued*

a. Define what a "hacker" is and distinguish between a "hacker" and a "cracker".
b. Give THREE ways to secure information on a network.
c. Why would it be valuable for hackers to steal research data and documents?
d. What are the consequences of being a victim of a ransomware attack?

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 A private network that is only available to the organisation`s employees. (1)
A. Internet **B.** WLAN
C. Intranet **D.** LAN

1.2 A network of computers covering a small area such as a school or office building. (1)
A. WAN **B.** LAN
C. GAN **D.** PAN

1.3 All the computers that are connected to the network. (1)
A. Work stations **B.** Servers
C. Input devices **D.** Network software

1.4 A network that uses radio waves to communicate rather than cable. (1)
A. WAN **B.** LAN
C. GAN **D.** WLAN

1.5 A computer that provides shared resources, such as files and printers. (1)
A. NIC **B.** Server
C. Workstation **D.** Network security

QUESTION 2: TRUE OR FALSE

Choose the answer and write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

a. If a network is not secured properly, the <u>entire</u> network will be infected should there be one infected computer. (1)
b. A <u>PAN</u> (personal area network) is designed to connect networks across large distances, for instance a province. (1)
c. In a LAN, networks are <u>globally</u> controlled, for easy changes and updates. (1)
d. An NIC (network interface card) allows a computer to communicate with a network. (1)
e. The main function of a <u>workstation</u> is to serve the information stored on it to the other connected computers. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from COLUMN B that matches a description in COLUMN A. Write only the letter next to the question number (e.g. 1-J). (5)

COLUMN A	COLUMN B
1. To make communication possible between the computer and the network.	**A.** Switch **B.** File server
2. Data transmission is slower when using this network.	**C.** Network security
3. A server that controls all the incoming and outgoing emails.	**D.** Email server **E.** LAN
4. A hardware device used to connect computers in a network.	**F.** WLAN
5. Policies that are in place to ensure the security of a network.	**G.** Network software **H.** NIC **I.** Hub

… continued

REVISION ACTIVITY

… continued

QUESTION 4: MEDIUM QUESTIONS

4.1 Besides computers, name one other hardware device you might find connected to a network. (1)
4.2 Give two reasons why a wireless network (WLAN) is better than a cabled network. (2)
4.3 List two possible disadvantages of using a WLAN. (2)
4.4 Provide two examples of servers that can be used in a network. (2)
4.5 Briefly explain the following communication devices:
- **a.** Modem (2)
- **b.** Router (2)

QUESTION 5: SCENARIO-BASED QUESTIONS

Choco`s Chocolate Company sells chocolates to distributors as well as directly to the public. The company`s head office is in Cape Town with branches in Port Elizabeth, Durban and Johannesburg. Each branch has its own LAN, which communicates with head office.

5.1 The company uses a LAN in all its branches.
- **a.** Briefly explain what a LAN is. (2)
- **b.** Give two advantages of using a LAN. (2)
- **c.** Give two disadvantages of using a LAN. (2)

5.2 Some of the employees have decided to communicate with their clients using IM.
- **a.** Explain how IM works. (2)
- **b.** Provide two reasons why the employees should be careful when using IM. (2)

5.3 The company uses **FTP** for the electronic sharing of documents larger than 6 GB.
- **a.** Why is it a good idea that the company uses FTP? (2)
- **b.** Give two disadvantages of using FTP. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1	Describe local area networks (LANS)?		
2	Discuss the advantages and disadvantages of LANs?		
3	Describe wireless local area networks (WLANs)?		
4	Describe the basic components of a network?		
5	Explain various types of connections?		
6	Discuss the basic components of a network, including workstations, servers and network devices?		
7	Describe how network connection speed is measured on wired and wireless networks?		
8	Explain how wired and wireless network communication works?		
9	Discuss the differences between the internet and an intranet?		
10	Define passwords in terms of basic network security?		
11	Discuss what usernames are?		
12	Describe what access rights are in terms of networks on web services?		
13	Discuss the advantages and disadvantages of various forms of online communication?		

TERM 3

CHAPTER 8

SOCIAL IMPLICATIONS OF COMPUTER NETWORKS

CHAPTER OVERVIEW	
Unit 8.1	Social issues for networks
Unit 8.2	Network security
Unit 8.3	Databases and big data
Unit 8.4	Normal currency vs cryptocurrency

By the end of this chapter, you will be able to:

- Define unauthorised access to networks
- Describe the ethical use of networks
- Explain what acceptable use policies (AUPs) in schools are
- Define the social implications for network usage
- Discuss network security and BYOD environments
- Explain the privacy issues you can encounter on a network
- Discuss personal responsibility in relation to network security
- Define databases and big data
- Explain the differences between normal currency and cryptocurrency

INTRODUCTION

We live in a digital world where everything about us, from our holiday pictures to our names and ID numbers, is somewhere on a computer. Companies are storing more of their private information on networks and people are using cloud storage to save more of their personal information online. With the click of a button, you can find out more about strangers on the other side of the world than you ever could before.

This is why network security is more important than ever. In this world, where all the information you need about someone is so easily available, attackers want to get their hands on this information and use it.

8.1 Social issues for networks

UNAUTHORISED ACCESS

Unauthorised access on a network is when someone gains access to a network using someone else's credentials or through other illegal methods. Accessing a network, website, account or service that you do not have permission to access is illegal.

This is why having usernames and passwords and setting up access rights for routers and other connection devices are important. Access to secure data on the network should also be controlled.

Authentication works to protect your personal information from unwanted access. You can protect your computer by setting up a username and password that you will need to enter each time you start it up.

You can also set up authentication for your smartphone and tablet. New types of authentication, beyond usernames and passwords, have made it possible for you to easily secure your devices and access them yourself. New types of authentication include biometrics (fingerprint-unlock or face-unlock) screen-lock patterns and PINs.

In this unit, you will build on your knowledge of authentication by looking at what unauthorised network access is and its negative consequences, how networks must be used **ethically** and why AUPs (in schools and other environments) are drawn up. We will also discuss network safety and security issues, especially when it comes to BYOD environments and the privacy issues surrounding networks.

ETHICAL USE OF NETWORKS

Since networks are used by a wide range of people, they must be used responsibly and ethically. Anything negative that any user does on a network can reflect negatively on the company, school or organisation whose network they are using. This is why most organisations set up **acceptable use policies (AUPs)** for those people who will be using their networks, especially when it comes to what users can and cannot do on the internet.

The chances are very high that your school has an AUP that you had to read and then sign or acknowledge in some way. This section will look at the basics of AUPs for schools and what guidelines they should contain. The AUP is there to protect the users (and the organisation) when they use ICT facilities and when they are online.

ACCEPTABLE USE POLICIES OF SCHOOLS

In a school, an AUP can cover a wide range of computing devices and networks and is a contract between the user (learner) and the organisation (school) that outlines what the user can and cannot do on a particular network.

Below are some guidelines that should be in AUPs:

- A list of basic **netiquette** rules, including not sending **spam** or **hoax emails** and how users should communicate using email or social media websites
- What may or may not be accessed online using the school's ICT facilities. This could include restricting access to social media websites
- How much information users may download from the internet (for example, no **live streaming** or downloads larger than a certain file size)
- Guidelines on respecting **copyright**, intellectual property laws and privacy, as well as how to avoid **plagiarism**
- When and how portable storage devices can be used
- Restrictions on what software can be installed on the school's computing devices
- What to do if users find that they have become the victims of identity theft, cyber-stalking and cyber-bullying, and what to do if their devices become infected with viruses or malware
- Clear descriptions of what will happen to a user who breaks the rules outlined in the AUP

Each school will have its own AUP depending on the computing devices, services and access they provide.

Activity 8.1

Answer the following questions:

1. How does authentication work?
2. What is an acceptable use policy?
3. Compare your school's acceptable use policy with the list of guidelines given above. Make a suggestion on how to improve your school's AUP.

UNIT

8.2 Network security

All networks can be the target of an attack, especially those that connect to the internet. This is why it is extremely important to understand and implement basic network security. In this section you will look at basic things that you can use or do to keep the network secure, for example:

- **firewall**
- antivirus software
- use passwords and usernames where necessary
- understanding BYOD environments and applying all relevant safety requirements

The first and most important is security software for your devices. Firewalls are built into most routers and computers to stop unwanted traffic getting onto your network. Firewalls are basically the security guards of a network. Everything that comes into your network from the internet should be going through a firewall.

All the computers in a network should also have anti-virus software installed to protect them from being infected with malware, viruses and other harmful programs. Anti-virus software constantly scans the programs on a computer and destroys any threats.

Most organisations have network security policies that outline the rules that are in place to protect the network. One of the most common policies is to give staff a username that they can use to log into the network and have them choose a password. When you need to choose a password, there are a few things you must keep in mind:

- The longer your password, the more difficult it is to crack. Make sure your password is at least 10 characters long or use a **passphrase** that is at least 15 characters long.
- Make sure that your password is something you can remember without needing to write it down. If you need to write your password down, make sure that you keep it hidden and mixed in with other notes, so it is not clear which one is your password. Keep these notes away from your computer.
- Do not use your name, a family member's name or the name of a pet. A password that cannot be linked to your personal information is the best kind.
- Your password should not be something easy to guess, such as "password" or "12345".
- Make sure that your passwords are a mix of uppercase and lowercase letters, numbers and special characters.
- Make sure that your passwords do not follow a pattern on the keyboard (such as "asdfgj").
- Never give out your password to anyone and try not to share your account details with anyone.
- You are responsible for keeping your password safe. Anything that is done on a network using your password and username will be your responsibility. To make sure your passwords are safe, you should change them every two months or so and you should never use the same password for different websites and networks.

FIREWALLS

https://www.youtube.com/watch?v=0QyXMYh6qLo

2018 FACEBOOK DATA BREACH

Cambridge Analytica Scandal: https://www.wired.com/story/wired-facebook-cambridge-analytica-coverage/

BASIC COMPUTER SECURITY PRINCIPLES

https://www.youtube.com/watch?v=6p_q_Xp--Rs

Case study The 2018 Facebook data breach

Read the case study below and answer the questions that follow.

On the 25th of September 2018, Facebook announced that a security flaw in its website allowed hackers to access the personal data of roughly 90 million users of the social network. The attackers used a vulnerability in Facebook's "View As" tool, which allows users to see what their profile looks like to other people. By using this vulnerability, the attackers stole Facebook access tokens that they could use to take over almost 50 million user profiles.

An access token is a short line of code stored in your browser or device that keeps you logged into your account without you needing to log in each time you want to access the website that issued it.

Facebook quickly patched the bugs in the code that gave the attackers access and logged 90 million users out of their accounts (the 50 million that were compromised and 40 million that were potentially compromised) on the web and on their mobile devices. Users who were logged out were greeted with a message explaining what had happened when they logged back in and a link to more details about the breach. Facebook also temporarily disabled the "View As" feature while they worked to patch up the code.

According to later reports, Facebook security personnel first noticed that something was wrong when they spotted a spike in unusual activity on the 16th of September 2018 and began investigating. The vulnerability had been in place since July 2017, which means that attackers could potentially have had access to the accounts for a long time.

The attack also left vulnerable users' other accounts where they had used Facebook to login to those sites. Facebook advised users to change their passwords and use their accounts' security and login page to see where they had been logged in with their Facebook credentials.

This data breach came hot on the heels of another scandal involving Facebook and a company called Cambridge Analytica, where it was found that Cambridge Analytica used the data from 50 million Facebook accounts, kept the data and used it to influence the 2016 American Presidential Campaign.

1. Do you think that Facebook did the right thing by logging users out before informing them of the data breach?
2. How do you think Facebook's security team could have handled the vulnerabilities better?
3. Do you think that Facebook is doing everything it possibly can to protect its users' privacy following the two major scandals?
4. What advice would you give to users whose accounts were affected by the data breach?

NETWORK SECURITY

Any time a network becomes accessible, it becomes vulnerable to attack. This is especially true of networks that connect to the internet. Another security concern for networks is making sure that the devices that connect to them are not making them open to attacks.

When it comes to BYOD environments, this could be difficult, since the network administrator does not have full control over what is done with those devices when they are taken home at the end of the day,

This section therefore looks at network and safety issues related to BYOD and privacy, and how they can be prevented.

Something to know

Microsoft Intune is a cloud service from Microsoft that helps organisations to manage the mobile devices that employees use to access the organisation's data and applications, thereby protecting their mail and data.

NETWORK SECURITY AND BYOD

The main concern with personal devices that are used for work is how to keep the company's private data separate from a user's personal data and how to make sure that an employee does not accidentally share that private data. One way to avoid this is to set up a data loss prevention policy and using **data loss prevention** tools. Organisations can also use mobile device management software, such as Microsoft Intune, to manage the devices on their networks.

Another concern is that most mobile devices, such as smartphones and tablets, might not have anti-virus software installed, or, if there is anti-virus software, it is not up to date. This becomes a problem when users accidentally download applications that contain malware, **adware** or **spyware**. Mobile devices can also become infected with viruses. When these devices connect to the network, they can spread the infection, making the network unsecured.

The last issue with BYOD environments is that employers cannot control what employees install on their devices. Employees may be able to install games or other media that can become a distraction at work.

Because of these issues, companies that use a BYOD policy must set up a clear AUP for their staff regarding what they can and cannot do with their devices.

PRIVACY ISSUES

With so much of what you do every day taking place on computers and online, privacy and who has access to your information has become a major issue. Networks have become powerful tools to access, collect, store and share personal data. Very often this data is used to provide a company's customers with a better service, or to make it easier and quicker for your doctor to access your medical history, for example. But this data can also be used by cyber-criminals to commit fraud or to steal.

It is the network owner's responsibility to make sure that the data their network stores and shares is accessed legally and that this data is only used for what they say its use will be. It is also their responsibility to make sure that this data is secure and safe.

PERSONAL RESPONSIBILITY

While it is true that network administrators must make sure that their networks are secure, you also have a responsibility to make sure that you do not expose a network you are using to attacks.

You do this by keeping the following tips in mind when using a network:

- Make sure that your devices have up-to-date antivirus software installed and that you never click on suspicious links or reply with personal information to suspicious emails.
- Respect others' privacy and products. Do not download or share content that has been obtained illegally (such as pirated movies or music) or content that violates someone else's copyright.
- Be careful what your share about yourself on the internet.
- Follow the AUPs of any network you are using.

If you keep these tips in mind, you can help make the networks you use more secure for everyone else.

Activity 8.2

1. What are the main concerns with BYOD policies in the workplace?
2. How can companies avoid security and usage issues in a BYOD environment?
3. List two tips you should keep in mind when using a network.

UNIT 8.3 Databases and Big data

DATABASES

Networks must be protected to safeguard the personal information stored on them. This is especially true of networks that have databases, since these are where personal information is stored. Databases are often the targets of cybercrimes, so database security is extremely important.

The foundation of database security is the confidentiality, integrity and availability (CIA) triad:

- Confidentiality is the most important aspect of database security. People should be able to trust that any information about them that is stored is safe and secure. Databases should be encrypted to make sure that this is the case.
- Integrity is another important aspect of database security. Integrity means that the data can only be accessed by people who have the correct permissions to view it. This can be done by making sure that access to the data is controlled.
- Availability means that the databases should be up and running when users need to access them. This means that any **downtime** should be scheduled and if there is unexpected downtime, database administrators should check that this is not due to a security breach.

BIG DATA EXPLAINED

What is big data and how is it used? https://www.bernardmarr.com/default.asp?contentID=766

BIG DATA

Big data, a term that describes the massive amounts of data that are generated every day by every single person, plays a major role in privacy concerns. More and more companies are collecting this data and storing it on databases to use in marketing or product development. Big data is used in online services like:

- **Online banking:** Proper analysis of big data can help detect any and all the illegal activities that are being carried out, like the misuse of credit cards, misuse of debit cards, customer statistics alteration and money laundering.
- **Booking reservations:** Big data from several sources has helped travel agencies, hotels and the tourism industry better understand what customers are looking for and this has led to more direct reservations.
- **E-learning:** Big data that is being collected is related to the students, faculties, courses and results. This can provide insights to improve the effectiveness of educational institutes.
- **Social websites:** Users of social websites share photographs, personal data and make comments on posts by other users. This information is used by social websites to provides users with personalised content and also to assist advertisers to hyper-target users.

Companies also capture big data on consumer habits for targeted marketing. This has raised concerns about privacy because every time you click on a website, post on social media, use a mobile app and comment via email or to call centres, your data is collected for future use. People have a right to their privacy but, without their knowledge or consent, this right is being eroded.

As big data increases, it exposes more of our data to potential security breaches. For example, if you have approved a company to analyse your data, how certain are you that they will not fall prey to a cyber-attack or that they will not sell your data. This could result in your private data being in unsafe hands.

Activity 8.3

In small groups discuss if and how big data has social implications for:

- Online banking
- Booking reservations
- E-learning
- Video conferencing
- Social websites

UNIT 8.4 Normal currency vs cryptocurrency

In the past few years, cryptocurrencies such as Bitcoin and Etherium have captured the general public's attention, but what is a cryptocurrency and how is it different from the normal currency you can get from a bank?

Simply put, cryptocurrency is a digital currency that can be bought, sold and traded online using cryptography. Cryptography takes the identifying data of the people buying and selling cryptocurrency and transforms it into something that is not readable unless it is deciphered. Cryptocurrencies are completely digital and have no physical form, unlike the money you get from the bank.

Normal currency is also usually backed by something physical that gives it value, like gold, and is usually issued and supported by a country's central authority or bank (in South Africa, that would be the Reserve Bank and the government). Normal, or fiat, currency is usually referred to as legal tender and has a value on the world's stock markets as well.

Cryptocurrencies are not backed by a single country's central authority and their value is not tied to anything physical. This means that they are decentralised and global.

Cryptocurrencies, however, are finite. There is only so much of it to go around.

Something to know

Bitcoin, the most popular cryptocurrency, was created in 2008 and remained mostly unknown. A few years later, Bitcoin mining became a popular pastime for computer enthusiasts and the popularity of Bitcoin grew. The value of Bitcoin in real world currency began to grow exponentially. In January 2017, a single Bitcoin would cost you $800 (about R11 104 in 2018). By the end of 2017, however, the price had skyrocketed until one Bitcoin cost $13 000 (about R180 000 in 2018)!

Activity 8.4

1. What does the term "Big data" mean?
2. Distinguish between normal currency and cryptocurrency.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following is NOT used to authenticate a user? (1)
- A. Username
- B. Password
- C. PIN
- D. BYOD

1.2 Which of the following techniques cannot be used to authenticate user access to your smartphone? (1)
- A. Fingerprint scanner
- B. Identity card
- C. Screen-lock pattern
- D. PIN

1.3 Which of the following scenarios will need an AUP? (1)
- A. A person wants to sell goods online.
- B. A person wants to open an internet café.
- C. A person buys car insurance.
- D. A person browses an online website.

… continued

REVISION ACTIVITY

... continued

1.4 What does BYOD mean? (1)
- A. Bring your own device
- B. Buy your online device
- C. Bring your device
- D. Bypass online detection

1.5 What is the purpose of a firewall? (1)
- A. It removes malware from your computer.
- B. It uninstalls programs from your computer.
- C. It updates your computer's software.
- D. It prevents outside devices or programs from accessing your network.

QUESTION 2: TRUE OR FALSE

2. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
   - a. <u>Authentication</u> is the process of working out whether something or someone is, in fact, what or who they claim to be. (1)
   - b. Using a network in an <u>unethical</u> manner can improve the reputation of an institution. (1)
   - c. On a network you must protect the <u>privacy</u> of others and yourself. (1)
   - d. Unauthorised access on a network is when someone <u>legally</u> gains access with another user's password. (1)
   - e. There are two main types of unauthorised access, internal and external <u>threats</u>. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. Encrypting people's personal information to ensure that it stays safe and secure.	A. Integrity B. Privacy issue
2. Controlling user access by making sure that data stored online can only be viewed by those who have the correct permissions to view it.	C. Password protection D. Availability E. Big data
3. Ensuring that users are able to have constant access to their online data.	F. Confidentiality G. Authentication
4. A digital currency that can be bought, sold and traded online using cryptography.	H. Cryptocurrency
5. Matching the password that a person provides with the credentials stored on a database to prevent the unauthorised access of data and information.	

QUESTION 4: MEDIUM QUESTIONS

4.1 Name three types of institutions that would benefit from AUPs. (3)

4.2 List three AUP guidelines that should be used by schools. (3)

4.3 Name three advantages of cryptocurrencies over normal currencies. (3)

... continued

REVISION ACTIVITY

… continued

QUESTION 5: SCENARIO-BASED QUESTIONS

Stacy has recently had a friend of hers install a network in her house. Since she has a four-year-old daughter and a 14-year-old son who will eventually have access to this network, she needs to work out what safety precautions she must implement.

5.1 What type of network should Stacy install? Give a reason for your answer. (2)

5.2 What three things can Stacy do to keep her network secure? (3)

5.3 What two things can Stacy do to control what her children can access on their home's network? (2)

5.4 Stacy's son has created a Facebook account. What three safety tips can Stacy give him about his use of this online platform. (3)

5.5 Stacy's son comes to her and tells her that he thinks that someone has hacked into his Facebook account. What three things should they do? (3)

5.6 Give three guidelines that Stacy can use when choosing a password that will be difficult to crack. (3)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Define unauthorised access to networks?		
2.	Describe the ethical use of networks?		
3.	Explain what acceptable use policies (AUPs) in schools are?		
4.	Define the social implications for network usage?		
5.	Discuss network security and BYOD environments?		
6.	Explain the privacy issues you can encounter on a network?		
7.	Discuss personal responsibility in relation to network security?		
8.	Define databases and big data?		
9.	Explain the differences between normal currency and cryptocurrency?		

TERM 3

CHAPTER 9

ERRORS AND BUGS

CHAPTER OVERVIEW	
Unit 9.1	Human error
Unit 9.2	Verification and validation of data
Unit 9.3	Software bugs
Unit 9.4	Hardware failure

By the end of this chapter, you will be able to:

- Discuss the effects of computer and human error on data accuracy
- Describe the garbage in, garbage out (GIGO) principle
- Explain the different data types
- Describe databases
- Describe how data is verified and validated
- Explain what software bugs are
- Define hardware bugs

INTRODUCTION

There are two major factors that can lead to a computer giving you the incorrect results when you enter data, namely human error and bugs. In this chapter, you will learn about the effect that human error has on data input and the accuracy of data, how data is verified and validated, and what software bugs and hardware failure are and how they can affect your computer.

UNIT

9.1 Human error

When data is added manually to a spreadsheet or database, there is room for error.

If the user makes a mistake when entering data, the computer will not be able to pick up on that. This is why we verify and validate data before analysing it.

Human errors can have a significant impact on the validity and accuracy of data. If, for example, a scientist makes a calculation mistake with one entry in a spreadsheet, their results could be off by a large degree. These types of errors can also impact businesses. If an employee enters the price of a product into the database incorrectly, the business may lose money.

THE GIGO PRINCIPLE

GIGO is a concept in data capture and evaluation. It stands for "garbage in, garbage out" and the idea is that if you enter bad data into a system (such as a database or Excel spreadsheet), you will get bad results, or bad input will give you bad output.

UNIT

9.2 Verification and validation of data

DATABASES

As you learned in Term 1, a database is a collection of a large amount of data, which is organised into files called tables.

When discussing data, you need to remember that there are many different types of data. The most common data types, and the ones you are most likely to deal with, are:

- strings
- numeric data
- boolean data
- date and time data.

VERIFICATION AND VALIDATION

In order to get the best results with your data, you will need to make sure that it is accurate. To do this, you will need to first verify the data and then validate it.

Data verification is the process of checking that the data a user has inputted is correct. This can be largely automated, providing someone has set up the rules on the spreadsheet or database. For example, the age range of high-school students is usually 14 to 18 and the spreadsheet can be set up to only accept an age in that range. This is called a range check. This does not always mean that the data will be 100% correct though. If a student is 15 and the user types in 17, the data is still valid in the range, it is just not correct.

Data verification is mainly used when data is entered into a system manually (that is, a human has entered the data into the database or spreadsheet) and there is a possibility that there will be errors in the data.

Data validation, on the other hand, is the process of making sure that the data that has been transferred from one source to the other matches the original data. For example, if you entered the results of a survey into a spreadsheet, you will check that the results you have match the results you entered.

Data validation also checks to see that the data is complete and matches the requirements of the system in which you entered it. In Access, this is done using data validation rules and input masks.

There are several ways to validate data. These are shown in Table 9.1.

Table 9.1: *Data validation techniques*

TYPE	HOW IT WORKS	EXAMPLE
Check digit	The check digit is used to check that the data entered is correct.	Barcodes have check digits at the end. They are calculated using the digits before them.
Format check	Checks that the data is entered in the correct format.	The correct way to enter a date in the spreadsheet is YYYY-MM-DD, so if a user enters the date as 21-10-2017, the spreadsheet will highlight it.
Length check	This makes sure the data is not too long or too short.	South African telephone numbers have 10 digits in them. If a user enters a telephone number with 11 digits, the spreadsheet will highlight that cell.
Lookup table	Looks up acceptable data in a table.	There are only 12 months in a year.
Presence check	Checks that there are no blank fields.	If you need the names of all employees, then a name field cannot be left blank.
Range check	Checks that a value falls within a specific range.	The number of subjects a learner has must be more than six and fewer than 10.
Spell check	Checks spelling and grammar.	This is most often found in word-processing software.

UNIT 9.3 Software bugs

A software bug is any error, flaw, failure or fault in a computer program or system that causes it to produce an incorrect or unexpected result or to behave in a way that it was not intended to. This can often cause the program to crash (or stop working) or produce an invalid response.

Did you know

Software bugs are called bugs because the first known programming flaw was caused by a moth that flew into a Harvard Mark II computer in 1946.

Most bugs are caused by human errors made when the source code for the program was written. Some bugs might not have a serious effect on how the program functions and may not be found for a long time, but a program may also be called "buggy" when it has several bugs that make it almost unusable. In some serious cases, programs may have bugs creating security flaws that can lead to a computer being accessed by cyber-criminals.

In the late 1990s, there was widespread fear that when the clock struck midnight on 1 January 2000, the Millennium Bug would cause software systems around the world to collapse, leading to an economic and social shutdown of the world.

This fear was caused because of a problem in the way that some early computers were programmed. They were only designed to handle years that contained two digits, so instead of using 1992, they would use '92. People started fearing that date-related processes would happen incorrectly for dates and times after 31 December 1999, since there was no way for the computer to tell the difference between the years 1900 and 2000, and that the computers would stop working when the date rolled over.

The idea that a simple change in the date could cause major computer systems to crash caused widespread panic due to the story being covered often in the media and being mentioned in reports on the topic from major corporations.

Needless to say, when the time came, no major computer failures happened. No one is sure if this was because many governments and companies upgraded their software or because there was nothing to fear in the first place.

UNIT

9.4 Hardware failures

Hardware failures are design errors in computer hardware that can cause the hardware to malfunction, fail or be damaged. Some hardware failures can also lead to security flaws in a computer's system.

FOR ENRICHMENT

Spectre and Meltdown are two hardware failures that affect CPUs built by Intel, AMD and ARM and can allow attackers to potentially steal sensitive data such as banking details and passwords. They were discovered in 2017 by a team from Google's Project Zero and several academic researchers from around the world.

Their results revealed that these failures had been in the hardware since about 1995 but that they had been previously undetected. They work by exploiting something called speculative execution (or the way a processor knows which task to fetch based on guessing what should happen next in the process and begins fetching instructions from where it thinks the program will go, without knowing for sure). This helps to speed up the computer's processors.

Combined, these two failures affect nearly every modern computer, including smartphones, tablets and PCs from different vendors running different operating systems. This is because the two failures are a fault in the processors themselves and are not linked to a flaw in any software.

While at the end of 2018 hackers had not released any software that could exploit these failures, computer and smartphone manufacturers advised that users should download and install the latest security fixes as they become available.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Before you analyse data, you must do the following: ______. (1)
- **A.** Spell check it
- **B.** Review it
- **C.** Verify and validate it
- **D.** Organise and simplify it

1.2 Information that is incorrectly inputted is referred to as which of the following? (1)
- **A.** Human error
- **B.** Transcription error
- **C.** Transposition error
- **D.** Troubleshooting error

1.3 Which of the following is an example of a transposition error? (1)
- **A.** Accidentally deleting an entry
- **B.** Repeating an entry
- **C.** Making a typo
- **D.** Swapping two entries around

1.4 Which of the following is NOT a data type? (1)
- **A.** Numbers
- **B.** Strings
- **C.** Dates and times
- **D.** Characters

1.5 What does GIGO stand for? (1)
- **A.** Generated Input Generated Output
- **B.** Good Input Garbage Output
- **C.** Good Input Generates Output
- **D.** Garbage In, Garbage Out

… continued

REVISION ACTIVITY

… continued

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

a. When we enter data into a computer, the errors that occur are usually computer-based. (1)
b. Transcription errors are most common with text-based data. (1)
c. Human errors negatively impact the validity and accuracy of data. (1)
d. Entering data correctly leads to incorrect results. (1)
e. Binary code is similar to date and time data. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. A process that is mainly used to check data that is entered into a system manually.	A. Data validation B. Boolean data
2. A file that can be used to make calculations on numerical data.	C. String data D. Numeric data
3. Basic text that is made up of a combination of characters, letters and numbers.	E. Transcription error F. Spreadsheet
4. Data that represents a count or measurements that can be used in mathematical calculations.	G. Hardware bug H. Data verification
5. Data can only have one of two possible values.	I. GIGO principle

QUESTION 4: CATEGORISATION QUESTIONS

4.1 Which data validation technique can be used to validate the data in the following scenarios? (7)

SCENARIO	DATA VALIDATION TECHNIQUE
1. An antivirus wants to confirm if its product key follows a specific pattern, before checking if it is correct.	
2. Checking the correctness of the numbers in a Fibonacci sequence.	
3. When you want to check if a person has provided both a username and a password on a login screen.	
4. When you want to check that the required fields on a form are completed.	
5. On a questionnaire you want people to only provide "Yes" or "No" answers.	
6. When you want to make sure that only users between 18 and 25 can open a bank account.	
7. When a bank wants to make sure that the user interface of an ATM makes it obvious where a PIN number and deposit amount must be entered.	

… continued

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REVISION ACTIVITY

... continued

4.2 Define a software bug. (2)
4.3 Define a hardware bug. (2)
4.4 Determine if the following errors are caused by software bugs or hardware bugs. (4)

ERROR DESCRIPTION	TYPE OF BUG
1. When Roberta plays her favourite role-playing game on her computer, her computer starts to freeze during the game's cut scenes.	
2. Nomonde's Windows computer is stuck at the Blue Screen of Death and is preventing her from logging in.	
3. Nothing happens when Prakash presses the "P" key on his keyboard, but every other key on his keyboard still works properly.	
4. Shou's laptop heats up very quickly and only stays on for an hour, even when it's plugged into a power source.	

QUESTION 5: SHORT AND MEDIUM QUESTIONS

5.1 What is the main difference between data verification and data validation? (2)
5.2 How can you automate the data verification process on a spreadsheet? (2)
5.3 Describe the two types of data entry errors and give an example of each. (4)
5.4 Maryn's mouse cannot be detected by her computer when she plugs it in. Mention two things she can do to try and solve this hardware bug. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Discuss the effects of computer and human error on data accuracy?		
2.	Describe the garbage in, garbage out (GIGO) principle?		
3.	Explain the different data types?		
4.	Describe databases?		
5.	Describe how data is verified and validated?		
6.	Explain what software bugs are?		
7.	Define hardware bugs?		

TERM 3

CHAPTER 10

SOCIAL ISSUES AND ONLINE PROTECTION

CHAPTER OVERVIEW	
Unit 10.1	Social-engineering tricks
Unit 10.2	Data protection
Unit 10.3	Protecting yourself online
Unit 10.4	E-commerce and e-banking
Unit 10.5	How antivirus programs work

By the end of this chapter, you will be able to:

- Identify different social-engineering tricks
- Describe why data backup is important
- Back up data correctly
- Discuss how to avoid online harassment and what to do if you become a target of online harassment
- Describe what malware is and how to avoid it
- Explain multi-step verification
- Describe how antivirus and antimalware programs work

INTRODUCTION

The internet is one of the most amazing inventions of our modern age. At the click of a button, you have access to millions of people across the world. Using a simple search, you can access information. You can learn almost anything on the internet and can teach yourself everything from simple activities such as cooking to more complex things such as making your own furniture.

But there is also a dark side to the internet. Digital crimes are carried out by criminals like hackers who want to steal your information and data, and cyberbullies and stalkers who harass people online. Knowing how to protect yourself from these criminals and crimes is an important part of your online life.

In this chapter, you are going to look at what **social engineering** is and learn about social-engineering tricks. You will learn about online harassment and how to avoid **cyberbullying** and **cyberstalking**. There is also advice on how to shop and bank online safely.

There is a section that discusses what antivirus software is and how it works, and you will also learn how to back up your data for your safety.

UNIT

10.1 Social-engineering tricks

HOW HACKING WORKS

This is how a hacker hacks you using simple social engineering:

https://www.youtube.com/watch?v=lc7scxvKQOo

SMS PHISHING SCAMS

Social-engineering techniques:

Phishing for email: https://www.youtube.com/watch?v=PnHT7ANA5vM

Social engineering is a technique used to gain access to a facility's systems, data and anything else by exploiting basic human psychology.

Some of these tricks include:

- **Phishing** is usually done via email where an attacker tricks people into giving out personal information or handing over money. Phishing is used with **email spoofing** and **website spoofing**. The most common phishing scam is an email from your bank to say your account has been suspended and that you need to reset the password by clicking on a link that takes you to a website that looks like your bank's website. When you enter your account details, the attacker will have your information.
- **Phishing and email spoofing:** Phishing and email spoofing attacks try to obtain sensitive information (such as usernames, passwords and banking details) by sending emails to users that look like official emails. These emails will either directly request the sensitive information, or redirect users to an official-looking website from where their information will be stolen.
- **Pharming:** Much like phishing, pharming attacks create an official-looking website that requests sensitive information. A very common pharming attack allows users to "change" their passwords. Instead of changing their passwords, the user's username and password are recorded and their account is taken over.

According to security studies, social engineering is the single most effective way to gain illegal access to a desired computer.

WHAT TO DO WHEN YOU SUSPECT YOU ARE A VICTIM?

If you think you accidentally revealed any financial information, immediately contact your bank and have them put a hold on any accounts that may be compromised.

Immediately change your password if you think you might have revealed it.

Activity 10.1

Do the following activity on your own.

1. Read through the following case study:

> Kevin Mitnick is known as the world's most famous hacker. He was one of the first computer hackers to be prosecuted and labelled as a computer terrorist, after leading the FBI on a three-year manhunt for breaking into computer networks and stealing software from Sun, Novell and Motorola.
>
> Mitnick started hacking when he was only 16 or 17 years old and later became known more for his use of social engineering to get access into networks than actually hacking them. For example, he called an employee of Motorola and convinced her to send him the code for one of the Motorola cell phones. With this, he was able to use an elaborate social-engineering scheme by manipulating the telephone network and setting up call-back numbers within Motorola's campus. He even convinced a manager in operations to tell one of the employees to read off his SecurID code any time he needed it, so that he could access the network remotely.

… continued

Activity 10.1

He also managed to hack into their development servers for cell phones and find the source code to all the different cell phones.

When Mitnick was eventually arrested in 1995, he was held for four-and-a-half years without a trial. In the end, he signed a deal and admitted to causing between $5 million and $10 million in losses to Motorola and other companies, although he kept saying that the purpose of his hacking was never for personal gain – he did it for the fun of being able to do it.

After he was released in 2002, he became a security consultant who is now doing exactly what he did as a hacker, i.e. breaking into computer and network systems, but with the company's knowledge and authorisation.

2. Scan the QR code on the left to watch an interview with Kevin Mitnick.
3. Discuss the following questions in groups:
   a. What is social engineering?
   b. Discuss four of the types of social-engineering techniques.
      i. What is it?
      ii. How is it used?
      iii. Have you or anyone you know been targeted by social-engineering tricks? What happened?
      iv. What can you do to prevent yourself from becoming a victim?

MITNICK INTERVIEW

https://www.youtube.com/watch?v=c76ezYglN28

UNIT

10.2 Data protection

It is important to back up your data regularly, since hardware can fail or accounts can be compromised. Backing up data lets you protect and store information.

CREATING A BACKUP

Because a back up is simply a copy of your computer's data, it can be created using the *Copy* and *Paste* functions.

Guided Activity 10.1

To do this, you should:

1. Start by setting up a backup schedule to work out how often you will create a backup.
2. Buy an external hard drive to use for the backup.
3. On the scheduled backup day, connect the external hard drive to your computer.
4. Select the files or folders you would like to back up and copy them using the *Copy* command.
5. Create a folder on the external hard drive with the correct date in the name.
6. *Paste* all the copied files and folders into this folder.
7. Disconnect the external hard drive and store it in a safe location.
8. Repeat these steps on each scheduled backup day.

To recover the files from this backup, simply connect the backup hard drive to your computer and copy the damaged or missing data back onto your storage device.

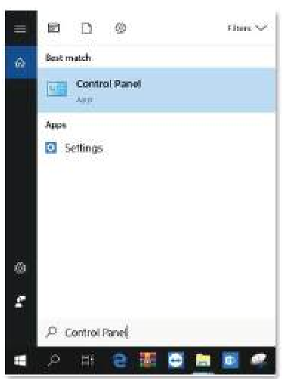
The problem with using *Copy* and *Paste* to back up your data is that the backups must be done manually. As a result, it is possible to forget to create backups. These backups will also be much larger, since each backup will contain all the files and folders you copied, even if some of these folders have not changed since the previous backup.

These problems can be solved by using a backup utility such as Windows 10's *Backup* and *Restore*.

Guided Activity 10.2

To use *Backup* and *Restore*:

1. Buy an external hard drive to use for the backup and connect it to your computer.
2. Open the *Start* menu and enter the words *Control Panel*. This will open the window.

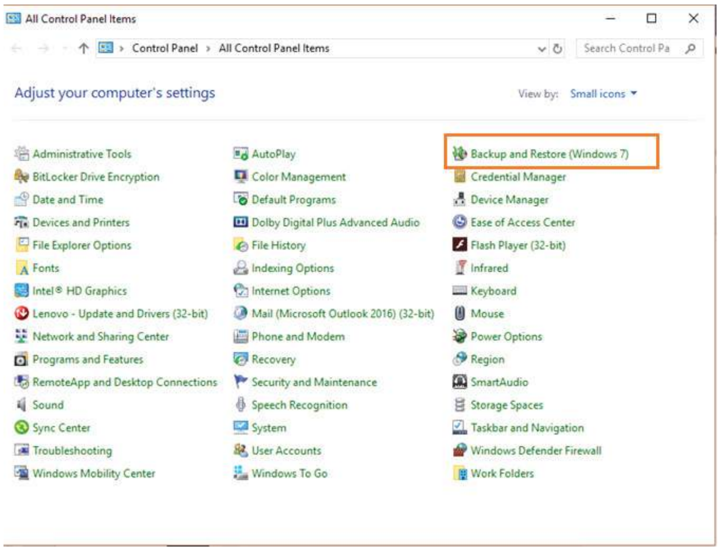


… continued

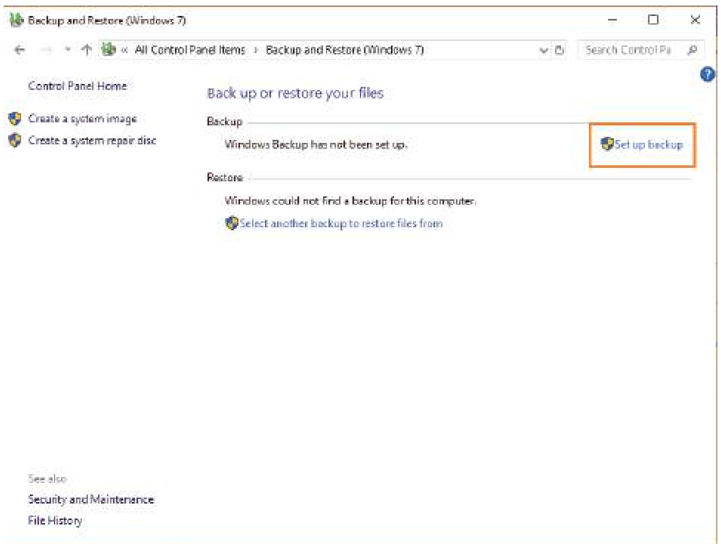
Guided Activity 10.2

… continued

3. Click on the *Backup and Restore* (Windows 7) option to open the *Backup and Restore* window.



4. Click on the *Set Up Backup* option.



5. Select the external hard drive as the location where you would like to save the backup.
6. If all your files are stored in your private folders (such as the *Documents*, *Music* and *Videos* folders), use the *Let Windows Choose* option. Otherwise, select the *Let Me Choose* option and manually choose the folders that should be backed up.
7. Click on the *Save Settings and Run Backup* option to create a backup of all your files.

Guided Activity 10.3

To restore files from a Windows backup:

1. Connect your external hard drive to your computer.
2. Open the *Backup and Restore* (*Windows 7*) application from the *Control Panel*.
3. Click on the *Restore All Users' Files* option.
4. Click on the *Browse for Files or Browse for Folders* option.
5. Tick the *In the Original Location* box and press *Restore*.

This will restore your files to where they used to be, thus replacing all lost files.

USING CLOUD STORAGE FOR BACKUP

One of the problems with backing up your computer's contents to an external hard drive is that your computer and the drive can be lost due to theft, flood or fire. Cloud storage puts your information on an off-site server that never goes offline and that can be accessed from anywhere through the internet.

Cloud storage, however, is not the same as cloud backup. Although they both work on a similar basis and the result is the same – storing files online – there are completely different reasons for using them.

With cloud storage, you choose which files you want to store online and you send them to your online account. When you delete a file that you stored online, it is still in your cloud storage because it is not really tied to your computer anymore. Cloud backup is when you install a program on your computer and you tell it to keep certain files backed up online. A backup service will also upload any changes that are made to those files so that the current version is always online.

A backup server works for people or businesses that have a lot of files they want to keep backed up online, so that, should one of the computers suddenly stop working, the files can still be obtained from the cloud backup server.

These are a few cloud storage providers:

- Google Drive: This works very well with Google products and you get 15 GB for free, after which you have to pay a monthly fee for more storage space.
- Apple iCloud: This service works with all Apple devices, whether a Mac, iPhone, iPad or iPod. You get 5 GB for free but have to pay for more than that.
- Dropbox: It gives users 2 GB for free and lets you access your files from the Web, your desktop or mobile devices.
- Microsoft OneDrive: Integrates well with Microsoft's products such as SharePoint and Word. Everyone with a Microsoft account gets 5 GB free storage space, after which you need to subscribe and pay for more space.

Activity 10.2

Answer the following in your own words.

1. Is it really necessary to back up the content of your computer? Why or why not?
2. Talk to at least two people you know who have their own computers.
   a. Find out if they back up their information? Why or why not?
   b. What do they back up?
   c. How many times do they back up their information?
   d. How do they back up their information?
3. What would be a practical and workable solution for you to back up your information?

UNIT

10.3 Protecting yourself online

ONLINE HARASSMENT, STALKING AND BULLYING

Online harassment is a wide range of damaging behaviours such as hateful messages, **distributed denial-of-service (DDoS) attacks**, **defamation** and others. The goal of the harasser is to drive the target off the internet or punish them by publishing personal information, sending threats, or promoting harm.

Online harassment is especially prevalent in young adults; with people of colour and women more likely to be victims of serious harassment.

HOW DO YOU AVOID ONLINE HARASSMENT?

Limit the personal data you share on social media accounts and review your security or privacy settings regularly. Do not post your physical address, mobile number or email address (or make them publicly available as part of your profile).

Choose your passwords wisely, make them unique and strong. Change passwords regularly and do not use the same password for different sites. Make sure that the answers to your secret questions are hard to guess.

WHAT TO DO IF YOU ARE HARASSED?

Make a copy of everything. Take screenshots of the harassment and print them out. Keep a record of the websites where the harassment happened. Have your parents take a look at the posts. Report the bullying to the website and ask that they take the harassing content down. If necessary, take the evidence you have collected to the police to open a case.

If you are bullied or harassed, you can call Childline South Africa on 0800 055 555.

MALWARE (MALICIOUS WARE)

Malware is any software that is specifically designed to disrupt, damage or gain unauthorised access to a computer. Examples of malware are:

1. Viruses
2. Worms
3. Trojans
4. Scareware
5. Spyware
6. Adware
7. Ransomware

To prevent your computer from becoming infected with malware, computer experts suggest that you do the following:

- Do not open suspicious emails.
- Do not download suspicious programs or attachments.
- Keep your antivirus application up to date.
- Keep your other software up to date: Hackers can use software weaknesses or vulnerabilities to gain access to your computer.

ONLINE HARASSMENT

Here's what online harassment looks like:

https://www.youtube.com/watch?v=usclHi8jk2w

Something to know

A study that was done in 28 countries, including South Africa, in 2018, found that South Africa showed the highest **prevalence** of cyberbullying, with an increase of 24% from 2011. This could be attributed to the increased use of social media in South Africa.

CYBERBULLYING

Teens Talk: What works to stop cyberbullying?

https://cyberbullying.org/teens-talk-works-stop-cyberbullying

What is cyberbullying?

https://www.pacerteensagainstbullying.org/experiencing-bullying/cyber-bullying/

MULTI-STEP VERIFICATION

Two-factor authentication, or multi-step verification, prevents anyone from logging into your accounts using just your username and password. Instead, they need a second factor (which is usually a physical device such as your phone) to access your account.

Activity 10.3

1. Read the following case study and answer the questions that follow.

   > Thando is a young woman who has always had weight and confidence issues. After she decided to improve herself and started following an exercise and diet regime, her friends encouraged her to take photos of herself to show the world her new-found body and self-confidence. Thando decided to share her progress and photos with her friends and family. When she uploaded her new photos on Facebook, a lot of people liked them. Unfortunately one person started to post horrible comments about her appearance and character.
   >
   > When Thando's sister saw these comments, she became worried that they would affect her sister's new-found confidence and her mental and physical health. She decided to talk to Thando about possible solutions to the problem. Both Thando and her sister eventually reported the person to Facebook and asked that Facebook remove that user's offensive comments and block their account.

   a. Have you experienced some kind of cyberbullying or harassment, or do you know of someone who has experienced it? Tell the story.
   b. Is it a problem in South Africa, or is it becoming a problem? Why do you say so?
   c. What can you as a young person do to prevent this from becoming a problem?
   d. What can you do if you learn that someone is being bullied?
   e. What must you do if you become the target of cyberbullying and harassment?
2. Is there something that you can do to stop cyberbullying at school? Come up with a school project.

UNIT

10.4 E-commerce and e-banking

When using online services such as online banking or shopping, you must be careful that your information is not stolen. Your banking or credit card information can be used to make multiple purchases.

Tips for safely using online banking and shopping include:

- Use proper antivirus software and make sure it is up to date.
- Do not log into online banking or shopping sites using public Wi-Fi or on public computers (like those in an internet café).
- Make sure that the operating systems on all your devices are up to date. This is important for both your phone and computer. Make sure that the banking app on your phone is also up to date.
- Change your passwords regularly and make sure they are strong.
- Do not sign into your banking account through a link in an email as this is most likely a phishing attack.
- Sign up for notifications on your bank account for when money comes in or goes out and when your account is accessed online.

You should always check that the website URL has an "https" extension whenever you are making a transaction online. The "s" stands for secure and it means that the website has something called a **Secure Sockets Layer (SSL)** certificate. This creates an encrypted link between you and the server, making the data that you and the server send and receive secure and private. This protects your data from being intercepted and stolen.

ADVANTAGES OF E-COMMERCE

Below are some of the advantages that e-commerce holds for the consumer, which is you.

- There is a very wide range of products and services available.
- There are no geographical limitations. It is just as easy to buy something from the USA or Europe as it is to buy it in South Africa.
- It definitely saves you time, for example travel time, waiting time and searching time.
- It is available 24/7.

RISKS OF E-COMMERCE

Unfortunately, there are also certain risks related to e-commerce, such as:

- Fraud: Because transactional data is transmitted over the internet, it has become one of the target areas of cybercriminals. Financial information can be hacked or stolen, leading to purchases made by people who are not the rightful owners of that information. Customers sometimes complain that they have not received their packages and it is difficult to determine whether that is true or not.
- Online security: There are many security threats such as malware, spam mail and phishing that can cause you harm.
- Exchanging a product that does not fit the online description can be difficult.

INTERNET BANKING

Internet banking explained:

https://www.youtube.com/watch?v=oADxUX4STjE

ADVANTAGES OF E-BANKING

Below are some of the advantages of internet banking:

- It is convenient, as customers have 24-hour access to the bank, seven days a week.
- It can be done from any location where you have Internet.

RISKS OF E-BANKING

Although online banking is secure enough to use on a daily basis, you should know about these risks:

- Fraudulent transactions
- Online theft of your access ID/user ID or PIN/password could happen.
- Banking apps can be compromised.

Activity 10.4

Answer the following in your own words.

1. What exactly does e-commerce mean?
2. What are the advantages of e-commerce?
3. What are the risks of e-commerce?
4. What can you do to make sure that you are safe when shopping online?

UNIT

10.5 How antivirus programs work

Antivirus software is any computer program that is designed to scan for, isolate and delete any harmful program from your computer before it causes any damage, but you can do an active scan any time you think that your computing device has been infected with a virus.

Antivirus and antimalware programs check all the files on a computer, checking all changes and the memory for specific types of file change patterns.

APPLICATION PERMISSIONS

When you install an app on your smartphone or tablet, you will need to give it permission to access certain functions on your device. For example, when you install a navigation app (such as Google Maps), you will first be asked for permission to access the device's GPS location.

Using app permissions means that you can control what capabilities or information the app can access.

Activity 10.5

1. Give two examples of ways to avoid online harassment.
2. What is malware?
3. How do you avoid getting a malware infection?
4. What is multi-step verification?
5. Give five tips for safe online banking.
6. What does the "s" in "https" mean?
7. Briefly describe three common types of virus detection in antivirus programs.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following is NOT an example of social engineering? (1)
- A. Phishing
- B. Malware
- C. Tailgating
- D. Pretexting

1.2 If you come across fake news, what should you do? (1)
- A. Post it and share it
- B. Do more research on it
- C. Delete it from the internet
- D. Edit and try to fix it

1.3 Which of the following institutions is used in phishing scams the most? (1)
- A. Banks
- B. Universities
- C. Kindergartens
- D. Online shops

1.4 Which of the following scenarios do NOT use online banking? (1)
- A. Ahmed owns an electronics store and does have a card swipe machine.
- B. Chloe does not have a physical clothing store but she sells clothing through an online website.
- C. Blessing only likes shopping for games online.
- D. Hanno wants to send money to his mother overseas.

1.5 Which of the following is NOT an antivirus program/application? (1)
- A. Avast
- B. Norton
- C. Bitdefender
- D. OneDrive

… *continued*

REVISION ACTIVITY

... continued

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

a. <u>Multi-step verification</u> can be used to keep your email password secure. (1)
b. A disadvantage of <u>e-commerce</u> is that it is hard to send physical money online. (1)
c. Emails from an unknown source are known as <u>suspicious emails</u>. (1)
d. If you want to prevent people from accessing your online accounts, you must use a <u>strong password</u>. (1)
e. Hackers use Quid Pro Quo to steal <u>passwords</u>. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. Done via email where an attacker tricks people into giving out personal information or handing over money.	A. Adware
2. The art of lying to get sensitive data. This can be done in person or over the phone.	B. Quid Pro Quo
3. When the attacker offers something someone wants and tricks the user into giving away login data to particular websites.	C. Phishing
4. The promise of something, usually a thing or a benefit, in exchange for information.	D. Firewall
5. When an attacker walks into an access-controlled environment by following a person with legitimate access.	E. Pretexting
	F. Baiting
	G. Spam
	H. Tailgating

QUESTION 4: MEDIUM QUESTIONS

4.1 Briefly explain how to recover data from a backup? (2)
4.2 What are three disadvantages of using an external hard drive to back up your computer's data? (3)
4.3 What is a preferable alternative to physical computer backups? (1)
   a. Explain what this alternative backup does. (3)
   b. Who would benefit most from this type of alternative backup? Mention three of them. (3)
   c. What three things can you do to prevent your computer from getting infected by malware? (3)
4.4 What is the difference between e-commerce and e-banking? (4)
4.5 List two risks of online banking. (2)
4.6 Explain the concept of cloud storage. (2)
4.7 Provide two examples of cloud storage providers. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Identify different social-engineering tricks?		
2.	Describe why data backup is important?		
3.	Back up data correctly?		
4.	Discuss how to avoid online harassment and what to do if you become a target of online harassment?		
5.	Describe what malware is and how to avoid it?		
6.	Explain multi-step verification?		
7.	Describe how antivirus and antimalware programs work?		

TERM 3

CHAPTER 11

INTERNET AND THE WORLD WIDE WEB

CHAPTER OVERVIEW	
Unit 11.1	Internet communication
Unit 11.2	Online transactions
Unit 11.3	Internet of Things (IoT)
Unit 11.4	Fourth Industrial Revolution (4IR)
Unit 11.5	Social media
Unit 11.6	Internet access
Unit 11.7	Browser and email software
Unit 11.8	Usability of web pages and websites

By the end of this chapter, you will be able to:

- Describe and discuss what Voice over Internet Protocol (VoIP) and video conferencing are
- Discuss the advantages and disadvantages of VoIP and video conferencing
- Describe the different types of transactions you can perform online
- Define the Internet of Things (IoT)
- Explain the advantages and disadvantages of social media
- Discuss good and bad practices on social media platforms
- Describe the limitations of fixed internet access
- Discuss mobile internet access in relation to Wi-FI hotspots, WiMAX, Bluetooth and mobile internet
- Differentiate between different email applications
- Describe the key factors that define the usability of websites and how websites link to word processing and forms

INTRODUCTION

The internet has changed the way we communicate with each other and the way we interact with the world. More people are making calls and reading their news online than ever before and the number keeps on growing.

In this chapter, you will learn about various forms of online communication, how the internet can be used for transactions such as banking, shopping and make bookings, and what the Internet of Things (IoT) is. You will also look at the advantages, disadvantages and best practices of social media and the different forms of internet access that are available. In the final two units, you will look at browser and email software, as well as what key factors make a website user-friendly.

UNIT

11.1 Internet communication

This unit examines different types of digital communication, such as VoIP and video conferencing, and discusses the advantages, disadvantages and best practices related to each of them.

VOICE OVER INTERNET PROTOCOL (VoIP)

VoIP is the technology that converts your voice into a digital signal, allowing you to make a call directly from a computer, a VoIP phone, or other data-driven devices.

Figure 11.1: *Skype using a smartphone, tablet and computer*

VoIP has several key advantages over traditional telephone exchanges, for example:
- The set-up costs are lower.
- VoIP has all the same features as normal telephone networks, but also includes features such as call forwarding, call waiting, voicemail, three-way calling and more.
- VoIP services can be accessed by staff anywhere they need it and these services can be centrally controlled.

VoIP also has its disadvantages, such as:
- VoIP relies on you having internet.
- There are also security concerns around VoIP calls. The data packets themselves can be intercepted and tampered with, and users are vulnerable to identity theft when using online VoIP services.
- You should also make sure that your network prioritises your VoIP data packets.

VIDEO CONFERENCING

Video conferencing is linked to VoIP and is often offered as part of a VoIP contract. Video conferencing works by using an online platform to make or receive video calls. The most popular of these platforms is Skype, which is also a p opular VoIP platform.

Figure 11.2: *Video conferencing works by using an online platform*

Video conferencing has many benefits for businesses in particular, such as:

- Reduced travel costs.
- Screen- and file-sharing allow the people taking part in the call to see what others are working on, which is good for collaboration.
- Video conferencing is a visual communication medium, so you can see the facial expressions and body language of the people you are talking to.

A disadvantage of video conferencing is bad video or audio quality can make a video call very frustrating.

You should treat video calls as you would in-person conversations, so make sure to be polite and respectful and avoid talking over other people.

Activity 11.1

Answer the following questions in your own words.

1. Compare VoIP and video conferencing. Tabulate the information.
   a. What is each of the above?
   b. What is each one used for (main function)?
   c. What are the main advantages of each?
   d. What are the main disadvantages of each?
   e. Give an example of each.
2. What is the difference between VoIP and video conferencing?

UNIT
11.2 Online transactions

The most popular is the EFT (electronic funds transfer), which allows you to log onto a banking website (or smartphone app) and send money from your account to another account.

Another popular online transaction is shopping. Online shopping is convenient.

Most travel agencies, hotels and airlines have websites where you can book all you need for your trip. You can use the World Wide Web to search for specials. You can also use airline websites for online flight check-in before you get to the airport.

Buy tickets to see your favourite performer or sports team. It is more convenient to book online and pay well in advance. Take a look at the case study below to see how online transactions fit into our everyday lives.

SHOPPING AND MAKING BOOKINGS ONLINE

Harry wants to surprise his wife with tickets to see her favourite band perform. He checks the ticket website and sees that all the Johannesburg shows are sold out, but there are still tickets available in Cape Town. He buys the tickets and decides that they should make a short holiday of their trip. He then goes to a website that compares flight prices. Harry enters the dates for when he wants to fly down and return, and finds a result that matches his budget. He books two tickets for those flights. Then he realises they will need somewhere to stay. Harry goes to an accommodation comparison site and enters the dates to check for availability. He sees that a lovely hotel close to the concert venue is offering an early booking special, so he books a room for a week. He logs onto his banking app and transfers his booking fee and deposit to the hotel. He also takes the time to reserve a rental car for the week. Following this, Harry decides to buy his wife a birthday gift while he is online and visits some shopping websites to look for a purchase. All of this takes him an hour from the comfort of his couch.

… continued

SHOPPING AND MAKING BOOKINGS ONLINE

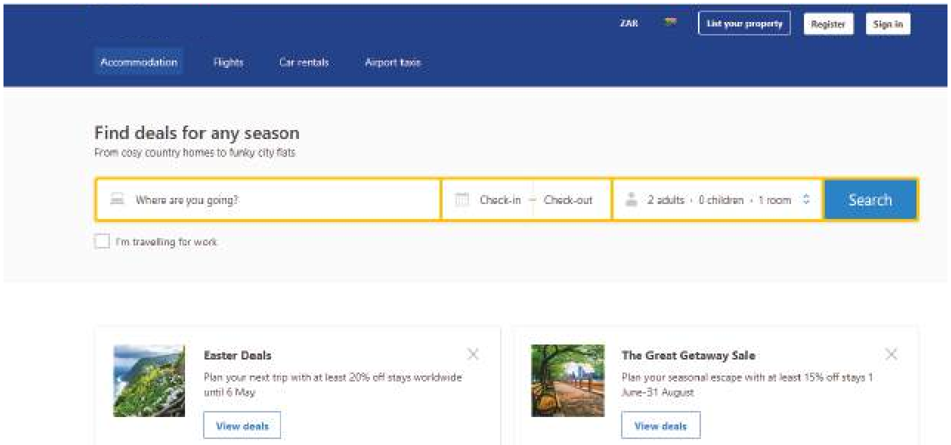


Figure 11.3: *Examples of online shopping and booking sites*

Activity 11.2

1. Why would a person use online banking for EFTs?
   a. Would you use it? Why or why not?
2. What are some of the drawbacks of online shopping?
   a. Have you done online shopping? Write a paragraph on your experience. Or explain why you do or do not like online shopping.
3. Give three examples of different online transactions (not banking or shopping).

UNIT

11.3 The Internet of Things (IoT)

THE INTERNET OF THINGS

More and more devices are able to access the internet. Various devices like fridges and washing machines have the potential to connect to each other through the internet. This has led to the rise of something called the Internet of Things, or IoT.

Everything that can be switched on and off can be connected in the IoT, from smartphones, coffee-makers and washing machines, to headphones, lamps and wearable devices. This also applies to components of machines, for example a jet engine of an airplane or the drill of an oil rig.

BENEFITS OF IoT

The IoT can be used to connect "smart homes", where you can use your smartphone to control everything from the temperature of the air-conditioners to the music that plays when you walk in the door.

The IoT can also be used to build "smart cities", where transportation and the movement of people can be controlled and monitored and made more efficient.

Figure 11.4: *Areas where IoT can be beneficial*

INTERNET OF THINGS EXPLAINED

https://www.youtube.com/watch?v=uEsKZGOxNKw

HOW IT WORKS: INTERNET OF THINGS

https://www.youtube.com/watch?v=QSIPNhOiMoE

Smart buildings can reduce energy costs by using sensors that detect how many occupants are in a room. The temperature can adjust automatically – for example, turning the air conditioner on if sensors detect a conference room is full or turning the heat down if everyone in the office has gone home.

In agriculture, IoT-based smart farming systems can help monitor, for instance, light, temperature, humidity and soil moisture of crop fields using connected sensors. IoT is also instrumental in automating irrigation systems.

In a smart city, IoT sensors and deployments, such as smart streetlights and smart meters, can help alleviate traffic, conserve energy, monitor and address environmental concerns, and improve sanitation.

Activity 11.3

Read the extract below and answer the questions that follow:

WITH SMART CITIES, YOUR EVERY STEP WILL BE RECORDED

By Sara Degli-Esposti & Siraj Ahmed Shaikh | 17 April 2018

Modern cities are brimming with objects that receive, collect and transmit data. This includes mobile phones but also objects actually embedded into our cities, such as traffic lights and air pollution stations. Even something as simple as a garbage bin can now be connected to the internet, meaning that it forms part of what is called the internet of things (IoT). A smart city collects the data from these digital objects, and uses it to create new products and services that make cities more liveable.

Although they have huge potential to make life better, the possibility of increasingly smarter cities also raises serious privacy concerns. Through sensors embedded into our cities, and the smartphones in our pockets, smart cities will have the power to constantly identify where people are, who they are meeting and even perhaps what they are doing.

Following revelations that 87 million people's Facebook data was allegedly breached and used to influence electoral voting behaviour, it is ever more important to properly scrutinise where our data goes and how it is used. Similarly, as more and more critical infrastructure falls victim to cyber-attacks, we need to consider that our cities are not only becoming smarter, they are also becoming more vulnerable to cyber-attacks.

Extract from https://theconversation.com/with-smart-cities-your-every-step-will-be-recorded-94527

1. What does the term Internet of Things (IoT) refer to?
2. Refer to paragraph 1. Many examples of IoT are given that could improve "liveability" in a city. Elaborate on any TWO of these examples (or give your own) by describing how they would improve citizens' lives.
3. How do the Internet of Things and Big data relate to each other?
4. Discuss two challenges a city would face when trying to implement the systems necessary to create smart cities, besides the challenge of funding such a project?
5. Many citizens of such smart cities say that privacy concerns are only relevant to people who have something to hide. Do you agree or disagree with this statement? Motivate your answer in a short paragraph.
6. Describe a scenario of the type of cyber-attack that could be launched on a smart city.

UNIT

11.4 Fourth Industrial Revolution (4IR)

Globalisation can be defined as a process of connection, interaction and integration among people, companies, and governments. This process is driven by international trade, investment, technology and big data. This has resulted in:

- An efficient market where there is an equilibrium between what buyers are willing to pay for a good or service and what sellers are willing to sell for a good or service.
- Increased competition between companies, which improves the service and quality of goods or services delivered to the consumers.
- Security as countries' economies are intertwined and dependant on each other.
- Wealth equality throughout the world as poorer nations have more job opportunities.

Society has gone through various stages. These are:

- The First Industrial Revolution, in the late 18th to early 19th centuries, is famous for industrialising agricultural work.
- The Second Industrial Revolution, in the late 19th and early 20th century, brought iron and steel into industry.
- The Third Industrial Revolution is the Digital Revolution with the age of the computer and the internet.
- The Fourth Industrial Revolution sees the digitisation of our society.

Globalisation and technology are intertwined as the movement of people, goods and ideas is accelerated and broadened by new forms of transport and communication. The spread of the internet and the relatively low cost of digital technology connect more people with the world. For example, small traders in shanties on the outskirts of Nairobi export across east Africa. In China, "Taobao villages" allow previously cut-off rural populations to sell goods on Alibaba's trading platform.

Sectors which the Fourth Industrial Revolution has impacted greatly includes:

- **Agricultural sector:** AI-powered machine vision systems can measure crop populations and detect weeds or plant pests and use robotic sprayers to precisely apply herbicides.
- **Healthcare sector:** Precision medicine helps doctors analyse a patient's genome sequence, medical history, and lifestyle, making a diagnosis more reliable.

THE FOURTH INDUSTRIAL REVOLUTION

https://m.youtube.com/watch?v=ChbjZHW2Wwk

Activity 11.4

In small groups, research the influences of the Fourth Industrial Revolution on the following sectors:

- Retail
- Building
- Social
- Travel

Activity 11.5

Answer the following questions in your own words.

1. Answer the following questions.
   a. Name three types of technologies from the 3IR that you have personally come into contact with.
   b. Name three types of technologies from the 4IR that you have personally come into contact with.

… continued

Activity 11.5 *… continued*

c. Do you think any technologies from the first three industrial revolutions can be replaced by technologies from the 4IR? Motivate your answer. Note: You can use an example to help you motivate your answer.

2. The internet is playing a large role in the lives of 60% of the total South African population, with 51% accessing it through their mobiles. Find out what the situation is in your school.
   - How many learners have access to the internet?
   - How many learners access the internet through their smartphones?
   - What brands of smartphones do they have?
   - How much time do they spend on their phones?
   - What are they doing on their phones?
   - What social media platforms are they using?
   - How does this compare to the world-wide trends?
     a. Create your own task definition using the knowledge you have gained in this section.
     b. Share it with the rest of the class.

UNIT

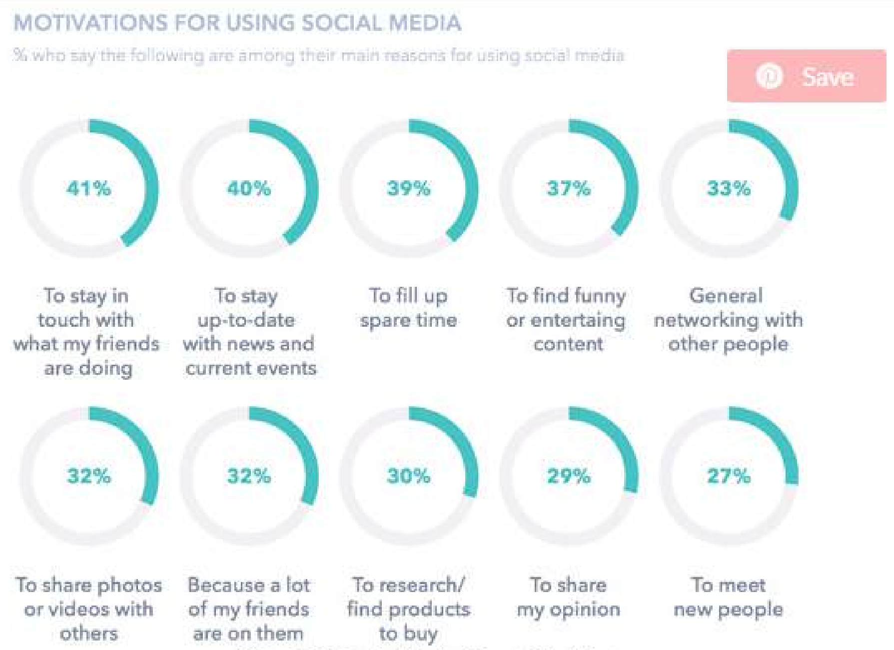
11.5 Social media

Social media has become a staple of our digital lives, from tweeting your thoughts on Twitter, to sharing pictures of your life on Instagram. Social media is used to market products, make announcements and stay in touch with the people who matter to you.

Something to know

Some interesting statistics on the use of social media

- There were 3,03 billion active social media users in the world in 2018.
- Facebook: 500 000 new users are added every day; that means six new profiles every second.
- Twitter: 500 million tweets are posted every day; this means 6 000 tweets every second.
- YouTube: 300 YouTube videos are uploaded every minute and 1 billion hours of videos are watched every day.
- Instagram: More than 95 million photos are uploaded every day.
- Pinterest: There are 200 million active users each month.
- Snapchat: There are 187 million active users every day.



Note: 2018 Social Media Use - Global Data
Source: https://www.globalwebindex.com/reports/social

How Participants Use Social Media Platforms In 2018 – GlobalWebIndex Chart

Figure 11.5: *The reasons why people used social media in 2018*

This section looks at the advantages and disadvantages of social media, good practices to use on social media and what to avoid doing when you are online.

ADVANTAGES

You can connect with anyone, anywhere in the world, staying in touch with friends who have moved away or family members who live in other countries.

Using social media keeps you up to date with what is happening in the world in real time. This is especially true of sites such as Twitter, where those on the ground at major events can tweet what is happening as it happens.

Social media platforms can be used to promote awareness and causes. Social activists use social media to inform people about issues and engage them in conversations about those issues.

Social media builds communities. Like-minded people can come together and discuss the things that interest or influence them.

DISADVANTAGES

Something called the social media bubble draws people to communities where their views (and only their views) are supported and encouraged. This can lead to people being misinformed about the world or only getting one opinion on an issue.

There is also a psychological effect on social media users. Since most people only share the positives in their lives on social media, others may compare their lives to the digital lives of their friends and family and find them lacking in something. Studies have shown that using social media can lead to depression.

Cyberbullying is also a big problem on social media and can cause great emotional harm to victims.

Social media is also used as a platform to scam or steal from people, especially older people who are not always internet literate and are more often likely to be the victims of 419 scams.

BEST PRACTICES

There are several good, common-sense practices you can follow when using social networking sites:

- **Manage your privacy settings:** Learn about and use the privacy and security settings on your social networking sites. They help you control who sees what you post and manage your online experience in a positive way.
- **Keep personal info personal:** Be careful how much personal information you provide on social networking sites. The more information you post, the easier it may be for someone to use that information to steal your identity, access your data, or commit other crimes such as stalking. Restrict who can have access to different types of information on your profile. Avoid publishing personal information like email addresses, home address and phone numbers.
- Think carefully about who you allow to become a "friend".

- **Restrict the amount of time you spend on social networking sites:** Many companies and schools block social media websites as a result of the abuse of resources like bandwidth, especially by downloading and uploading large files, photographs and videos.

Activity 11.6

Answer the following questions in your own words.

1. List three social media websites you use. Also explain why you use each of them.
2. Mention three advantages and three disadvantages of social media websites for you.
3. Mention three things you must do and three things you must not do when using social media. Also give the reasons why you think that.
4. Why should you investigate news articles before you share them?
5. Do you know of any examples of fake news? Discuss what you know about it and how it can affect people who read it.

UNIT

11.6 Internet access

Being able to access the internet allows you to access a world of information with a simple search and the click of a button. This unit will look at the limitations of fixed-line internet access and the different types of portable internet access and connections available, such as Wi-Fi hotspots, Bluetooth and mobile internet connections.

MOBILE INTERNET ACCESS

There are many types of mobile internet access, such as cellular data signal, Wi-Fi hotspots and WiMAX. This unit will look at several different types of mobile internet access and where each one is usually used.

WI-FI HOTSPOTS

A Wi-Fi hotspot is a wireless access point that you can connect to using the wireless connectors in your computing device. When we talk about Wi-Fi hotspots, we are usually talking about the publicly accessible connection points that businesses supply (usually for free) to their customers.

You can also create a wireless hotspot in your home by using a router with a wireless connector, or on the go by using your cellular phone or other devices capable of connecting to the internet.

Wi-Fi hotspots can usually be found in coffee shops, airports and hotels. Some Wi-Fi hotspots can also be provided as a public service, for example, the city of Tshwane in Gauteng offers a service called TshWiFi.

Your wireless capable device (laptop or smartphone) can search for and connect to Wi-Fi hotspots nearby.

WiMAX

WIMax is used to connect multiple devices over a longer range than Wi-Fi could and as a replacement for the GSM connections standards for mobile devices. WiMAX can cover long distances (like a cellphone signal network) and deliver high-speed internet access (like broadband connections).

More devices have Wi-Fi connectors built in than they do WiMAX connectors and most mobile carriers choose to use Long-Term Evolution (LTE) instead of WiMAX in their networks.

BLUETOOTH

Bluetooth is a wireless communication standard that allows electronic devices to connect to and interact with each other wirelessly. Bluetooth does not rely on mobile data, cellular signal or Wi-Fi to connect, as long as the devices that want to connect are within range of each other (and have each other's pass codes), they can connect.

MORE ON BLUETOOTH

Watch this video if you want to learn more about Bluetooth.

https://youtu.be/jzxZUJmOu30

Bluetooth can be used for many different things, like connecting Bluetooth-enabled devices such as keyboards, mice and speakers to your computer or headphones to your smartphone. You can also use it to send documents to Bluetooth-enabled printers or connect your smartphone to your car's Bluetooth system to make hands-free calls, listen to music or listen to text messages out loud and send them using text-to-speech input.

NFC

NFC (Near field communication) is a method of wireless data transfer that detects and then enables technology in close proximity to communicate without the need for an internet connection.

You can use NFC at a local supermarket, train station, taxi or coffee shop that supports contactless payments via your phone's NFC chip.

MOBILE INTERNET

Mobile internet (or mobile broadband) is the term used to describe the wireless internet access you can get through cell phone towers and other digital devices that use portable modems (such as tablets and smartphones).

Mobile internet lets smartphone users connect to the internet wherever they have decent network signal. Most cellular service providers also offer mobile internet packages where you use a small modem (called a dongle) to connect to the cellular network. These dongles can either plug directly into your computer (using USB) or can be used to create a Wi-Fi hotspot.

It is also an alternative to fixed-line internet connections (such as ADSL) and is useful in areas where ADSL services are not available or for users who do not want to have a telephone line installed.

There are different types of mobile internet connections. The most common are 3G (which stands for Third Generation) and LTE or 4G. LTE is the current standard for mobile internet. It is faster than 3G and can support things like high-definition video streaming. Looking to the future, 5G promises to be even faster than 4G and will become the new standard in internet connection for mobile devices.

With mobile internet, you can use your smartphone as a modem. You can turn your smartphone into a Wi-Fi dongle to connect other devices to the internet using something called tethering. Using your smartphone as a modem gives you access to the internet in areas where there are no Wi-Fi hotspots.

MORE ON 5G

Watch this video if you want to learn more about 5G.

https://m.youtube.com/watch?v=MC_Sfkh5-zQ

Figure 11.6 shows an example of the tethering screen on an Android device.

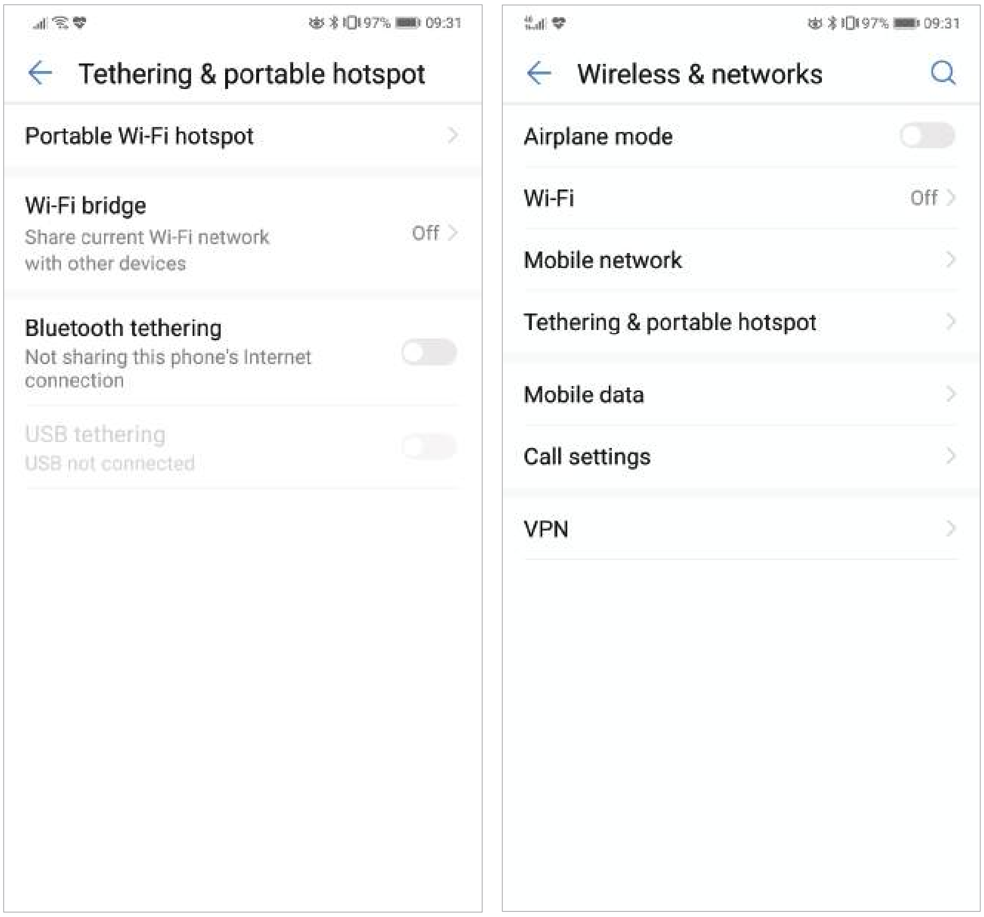


Figure 11.6: *Tethering settings on an Android device*

Activity 11.7

Answer the following questions in your own words.

1. Briefly discuss the limitations of fixed internet access, paying particular attention to the South African context.
2. What is a Wi-Fi hotspot and give an example of one?
3. What does WiMAX stand for?
4. What is Bluetooth and how does it work?
5. List the two most common types of mobile internet connections and describe how they differ from each other.

UNIT

11.7 Browser and email software

There is a huge range of internet browsers available for you to choose from. Browsers are simply the software you use to access the websites on the World Wide Web.

Some browsers are specifically linked to a manufacturer or operating system, such as Safari for Apple devices with iOS and MacOS, or Microsoft Edge for devices with Windows OS. This is not to say that you cannot download browser software from the internet and use it instead of the browser that comes standard with your device.

EMAIL

Email is a widely used form of communication. It consists of software for creating, sending, receiving and organising electronic mail (or email). Modern desktop email clients like Microsoft Outlook, Windows Live Mail and Mozilla Thunderbird offer advanced features for managing email, including WYSIWYG editors for composing email messages, anti-spam and anti-phishing security protection, advanced search capabilities, and rules and filters for more efficiently handling and organising messages and email folders.

A large number of online email services, called webmail, exist with features and functionality for managing email similar to their desktop email software counterparts. Some of the more popular online email services are Yahoo! Mail, Gmail, Hotmail (Windows Live Mail) and AOL Mail.

Figure 11.7: *The Gmail, Outlook and Apple Mail application logos*

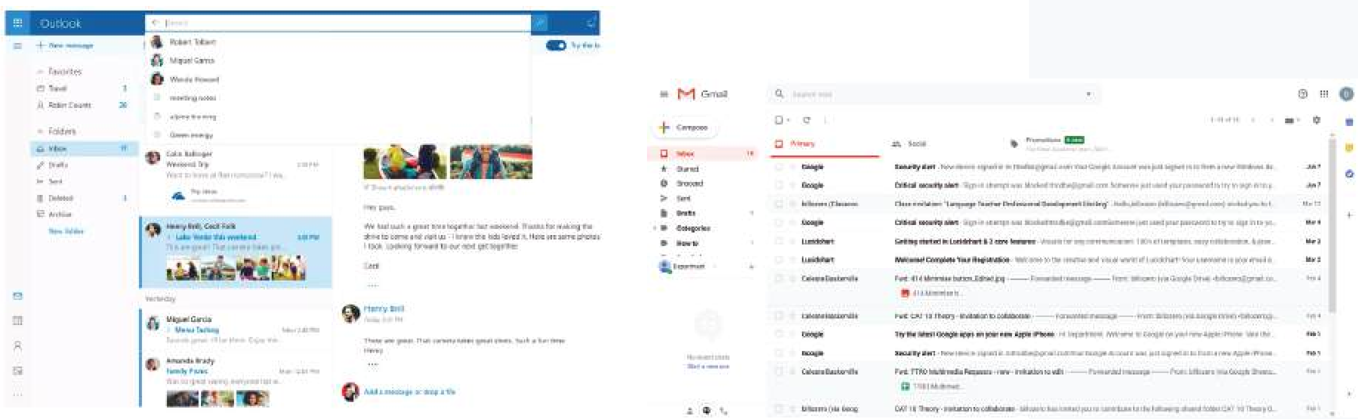


Figure 11.8: *Email user interfaces*

Gr11_CAT_Theory_LB.indb 151 2019/11/19 10:36 AM

UNIT

11.8 Usability of web pages and websites

Website usability looks at how user-friendly a website or web page is. There are five key factors that determine the usability of a site:

- **Readability:** This feature is one of the more important aspects of web design usability. Readable text affects how users process the information from the content. Poor readability scares readers away from the content. On the other hand, done correctly, readability allows users to efficiently read and process the information in the text. You want users to be able to read your content and absorb it easily.
- **Navigation:** Navigation is very important to ensure users are able to find what they are looking for easily and intuitively. The structure of the navigation should be simple and the main links (or menu system) should be easy to locate and identify. They should always appear in a consistent position on the website. Links should be short and it should be simple to figure out what the link is leading to. Web designers should ensure that no broken links occur on the site.
- **Consistency:** This feature affects both usability and readability. Consistency means, for example, that all headers of the same importance should be treated the same in terms of size, colour and font etc. For example, all <h1> headers in an article should look identical. This consistency provides users with a familiar focus point when they are scanning the text, and it helps to organise the content.
- **Layout:** Layout refers to how the various elements (text, graphics, buttons, etc.) are arranged on a web page. Pages should be designed and laid out in a way best suited to their intended audience or readers. Text and graphic objects on the page should be adapted to fit standard monitor sizes and resolutions.
- **Typography:** Typography refers to fonts and how they are put together. The font you use on your website needs to meet two specific criteria:
  1. How easy is it to read?
  2. Can it be rendered in HTML?

Each of these factors covers a single aspect of the website and they all need to work together to create a user-friendly website.

LINK TO WORD PROCESSING AND FORMS

Websites can link to word processors and word-processing forms in a few ways:

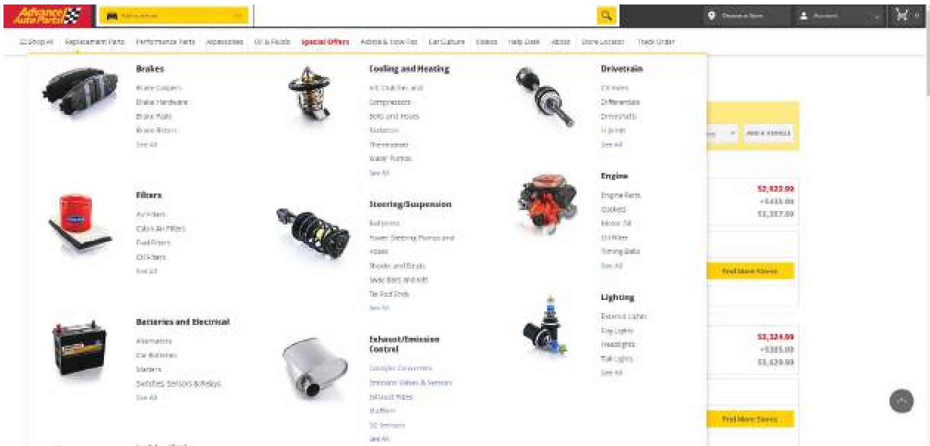
1. It is possible to create basic web pages in some word processors (such as Microsoft Word).
2. You can use word processors to create content for your websites.
3. You can use word processors to create forms for your websites.

Activity 11.8

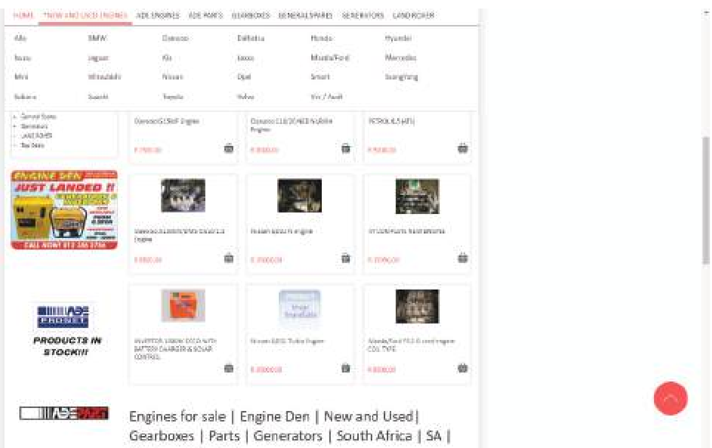
Answer the following questions in your own words.

1. Below are two websites that sell car engines. Look at the navigation designs of these websites and answer the following questions.

Website A



Website B



   a. Which website has the best readability? Give two reasons for your answer.
   b. Which website has the best navigation? Give two reasons for your answer.
   c. Which website has the best consistency? Give two reasons for your answer.
   d. Which website has the best layout? Give two reasons for your answer.
   e. Which website has the best typography? Give two reasons for your answer.
2. Based on your comparisons of the two websites, which website do you prefer. Give two reasons for your answer.
3. Name two things that can be done to improve the website you least prefer.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following cannot be done with VoIP? (1)
- **A.** Voicemail
- **B.** Call forwarding
- **C.** Call waiting
- **D.** Instant messaging

1.2 Which of the following is not a disadvantage of video conferencing? (1)
- **A.** Lagging caused by bad internet connections
- **B.** Technology can fail during a call
- **C.** Remote access
- **D.** Bad audio quality

1.3 What does EFT stand for? (1)
- **A.** Electronic Financial Transfer
- **B.** Electronic Funds Transfer
- **C.** Electric Finance Transfer
- **D.** Electric Fund Transfer

1.4 Which of the following is an example of e-commerce? (1)
- **A.** Online shopping
- **B.** Online banking
- **C.** Online government services
- **D.** Online gaming

1.5 Smart fridges use embedded ____________ to connect to the internet. (1)
- **A.** Network adapters
- **B.** Network cables
- **C.** Wi-Fi
- **D.** Interface cards

QUESTION 2: TRUE OR FALSE

Choose the answer and write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- **a.** VoIP converts <u>digital</u> signals to <u>analogue</u> signals. (1)
- **b.** VoIP can work with an <u>unstable</u> internet. (1)
- **c.** Data packets are <u>protected</u> from internet attacks. (1)
- **d.** The Internet of Things is a system in which everyday appliances connect to <u>computers</u>. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. A washing machine that can be turned on remotely.	**A.** Cyberbullying
2. A website where 1 billion hours of video are watched each day.	**B.** YouTube
3. An application that people can use to make video telephone calls.	**C.** Online banking
4. A technique used to pay for online purchases.	**D.** Internet of Things
5. Emotionally abusing people online.	**E.** Cybercrime
	F. Vimeo
	G. Big data
	H. VoIP
	I. EFT
	J. Facebook

… continued

REVISION ACTIVITY

… continued

QUESTION 4: FILL IN THE MISSING WORDS

Fill in the missing word(s) in the following statements. Provide only one word for each space.

a. EFTs allow you to log onto a ____________ website and send money from your account to another account. (1)
b. If you do not have physical money on you, you can make a payment by ____________. (1)
c. Any object that makes use of a(n) ____________ source can connect to the Internet of Things. (1)
d. ____________ is the most popular social network. (1)
e. Social media ____________ things and people together. (1)

QUESTION 5: CASE STUDY

Read the extract below and answer the questions that follow:

WITH SMART CITIES, YOUR EVERY STEP WILL BE RECORDED

By Sara Degli-Esposti & Siraj Ahmed Shaikh | 17 April 2018

Modern cities are brimming with objects that receive, collect and transmit data. This includes mobile phones but also objects actually embedded into our cities, such as traffic lights and air pollution stations. Even something as simple as a garbage bin can now be connected to the internet, meaning that it forms part of what is called the Internet of Things (IoT). A smart city collects the data from these digital objects, and uses it to create new products and services that make cities more liveable.

Although they have huge potential to make life better, the possibility of increasingly smarter cities also raises serious privacy concerns. Through sensors embedded into our cities, and the smartphones in our pockets, smart cities will have the power to constantly identify where people are, who they are meeting and even perhaps what they are doing.

Following revelations that 87 million people's Facebook data was allegedly breached and used to influence electoral voting behaviour, it is ever more important to properly scrutinise where our data goes and how it is used. Similarly, as more and more critical infrastructure falls victim to cyber-attacks, we need to consider that our cities are not only becoming smarter, they are also becoming more vulnerable to cyber-attacks.

Extract from https://theconversation.com/with-smart-cities-your-every-step-will-be-recorded-94527

5.1 What does the term Internet of Things (IoT) refer to? (2)
5.2 Refer to paragraph 1. Many examples of IoT are given that could improve "liveability" in a city. Elaborate on any TWO of these examples (or give your own) by describing how they would improve citizens' lives. (2)
5.3 How do the Internet of Things and big data relate to each other? (2)
5.4 Discuss two challenges a city would face when trying to implement the systems necessary to create smart cities, besides the challenge of funding such a project? (2)
5.5 Many citizens of such smart cities say that privacy concerns are only relevant to people who have something to hide. Do you agree or disagree with this statement? Motivate your answer in a short paragraph. (2)

QUESTION 6: SCENARIO-BASED QUESTIONS

Like many of his friends, Zike likes to stay up to date with the current affairs of the world, gossip and the media. Zike's teacher has noticed that a lot of his learners get distracted by their smartphones and as a result do not pay attention in class.

6.1 What three dangers are the learners most vulnerable to through social media websites such as Facebook? (3)

… continued

REVISION ACTIVITY

… continued

6.2 What four best practices should Zike's teacher suggest to the learners to help them navigate the social media they are regularly exposed to? (4)

6.3 Recently, Zike has started to notice that someone has been stalking him online.

- **a.** Mention one thing he can do to solve the problem.
- **b.** Mention one thing he should avoid to prevent the problem from getting worse. (2)

6.4 Briefly explain why Zike needs to know what data cap is. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Describe and discuss what VoIP and video conferencing are?		
2.	Discuss the advantages and disadvantages of VoIP and video conferencing?		
3.	Describe the different types of transactions you can perform online?		
4.	Define the Internet of Things?		
5.	Explain the advantages and disadvantages of social media?		
6.	Discuss good and bad practices on social media platforms?		
7.	Describe the limitations of fixed internet access?		
8.	Discuss mobile internet access in relation to Wi-FI hotspots, WiMAX, Bluetooth and mobile internet?		
9.	Differentiate between different email applications?		
10.	Describe the key factors that define the usability of websites and how websites link to word processing and forms?		